

# Machine Learning for Safety and Efficiency in the Naval Environment:

*Predictive Analysis of Naval Incidents in the USA, 2002 - 2015*

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Master's Thesis Report

*Big Data and Data Science applied  
to economics and administration  
and Business Management (UNED)*

December 2023



## Abstract

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This master's thesis focuses on the predictive analysis of naval accidents. It is based on a database of accidents recorded by the United States Coast Guard (USCG) from 2002 to 2015, along with meteorological data provided by the National Oceanic and Atmospheric Administration (NOAA). This database contains incident characteristics, as well as information about the involved vessels and recorded weather.

To achieve this, different techniques framed in data science will be applied. Specifically, by comparing different classification algorithms, the study aims to provide information and useful statistical tools that contribute to advancing the prevention of life, environmental, and financial risks in the maritime environment.



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## 1. Introduction

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The maritime industry has always played a vital role in the global economy. Due to the environment in which it operates, security has always been one of its main challenges.

Naval incidents have potentially catastrophic consequences on a vital, environmental and financial level. This is why its prevention has been a source of study since ancient civilizations such as Ancient Egypt, the Roman Empire, China, Ancient Greece or the Viking peoples.

Since then, naval records have been used to which successive mathematical, statistical and scientific advances have been applied to prevent incidents. Following this path, today you can count on computing. And getting to the tip of this iceberg, there's big data and data science. Therefore, paradoxically, this work aims to solve a very old question with the most developed methodologies and techniques so far.

By leveraging the wealth of data collected by the U.S. Coast Guard and supplementing it with weather information provided by the National Oceanic and Atmospheric Administration, this study seeks not only to analyze past incidents, but also to anticipate and mitigate future risks

Throughout these pages, the importance of collecting and processing relevant data from past incidents, as well as the implementation of advanced machine learning algorithms and data analytics to forecast potential incidents in the future, will be highlighted.

The data analysis will be carried out with the help of the Python language, using packages such as Numpy, Pandas, Caret, Keras and H2o. The choice of this tool is based on its specialized capacity for data analysis and the implementation of machine learning algorithms that are developed and tested by an active community. In addition, its ability to generate reproducible documentation using Jupyter Notebooks makes it a good choice to publish the results in a very clear way.

The transmission of information obtained in the analysis is instrumentalized by means of this report, as well as its corresponding annexes that contain the Python code used. Which are detailed below:

- Annex 3.1. Preprocess All Casualty & Pollution
- Annex 3.2. Preprocess Weather Ocean
- Annex 3.3. Preprocess Weather River
- Annex 3.4. Preprocess Merged Activity & VesselBalancedSample
  
- Annex 4.1. EDA VesselBalancedSample
- Annex 4.2. EDA MergedActivity
  
- Annex 5.0. Python Functions
  
- Annex 5.1. Data Model VesselBalancedSample
- Annex 5.2. Data Model MergedActivity

The numbering corresponds to the sections of this report and will be referenced when appropriate to facilitate its understanding.

## 2. Goals

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In this work, general goals will be pursued both in the subject of study (safety at sea) and in the field of data science.

On the one hand, it is intended to provide accurate information for the safety and efficiency of both crews and the assets used in the maritime environment. Specifically, the aim is to identify the conditions in which the most incidents have occurred during the study period, so that they can be extrapolated to the present and to other areas of the world through predictive models. Mainly, it is intended to answer these two questions for a given boat:

1. Can a vessel be involved in an incident?
2. And if so, what kind of incident?

The tools to answer these questions are going to be predictive classification models. For the first question, we are going to use a binary variable that classifies whether a ship is involved in an incident or not. And for the second question, it will be through a variable that includes the type of accident.

For this analysis, a high volume of data has been considered. In this way, the effectiveness of statistical and probabilistic approaches such as Bayesian models, neural networks, Gradient Boosting, etc., which will be discussed later, will be tested. It is intended that these models will be able to predict possible incidents with the input of new real data, so that the fit of each model with the most appropriate parameters will be maximized.

Regarding the operational objectives for the achievement of this work, we can point out the following points:

- Data collection. Data on naval incidents are publicly available on the [U.S. Coast Guard \(USCG\) website](#) and weather data are also available on the [U.S. Oceanic and Atmospheric Administration](#) (NOAA) website.
- Pre-processed. As will be seen later, there is an important work in this section, since data from three independent sources has had to be combined, as well as exhaustive cleaning and filtering to obtain the databases.
- Statistical exploration. This is a key chapter of any predictive analytics. But in this case, this is also one of the main goals of the analysis, as information from historical data is especially valuable in preventing future events.
- Modelling. It is one of the central motivations of this work, representing the bulk of the information exposed. Several algorithms have been implemented, comparing them with each other using the most common performance metrics in this type of analysis. To understand how they have worked, this section will discuss their explainability.
- Conclusions. As a consequence of the above aspects and taking into account that this information should be useful for people without in-depth knowledge of data science, the information obtained in the modeling has been synthesized.

### 3. Databases

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The databases used for this analysis come from three different sources, provided by U.S. government agencies:

- USCG: Marine Casualty & Pollution Data for Researchers ([link](#))
- NOAA: Global Marine data ([link](#))
- NOAA: Daily Summaries ([link](#))

The choice of this data is based on availability criteria. In the first instance, an attempt was made to obtain maritime safety data in Europe. But the custodian authority, EMSA, refused to share them for academic purposes, citing data protection issues. After a "cold door" search, they evaluated the data collected by the Transportation Safety Board of Canada for their country. However, when they were evaluated, the quality of their data was found to be unsatisfactory. Finally, it was the data published by the U.S. Coast Guard that met the conditions to meet the objectives set.

These datasets have undergone a reduction of information before being combined with each other in the two datasets used to answer each of the above questions:

1. **VesselBalancedSample**: Contains characteristics of ships involved in accidents, as well as those not involved. In this way, the aim is to extract the characteristics of the damaged boats.
2. **MergedActivity**: Contains information on both the incidents and the ships involved in them. The aim of this dataset is to extract the characteristics of each type of accident

For the aforementioned reduction of information, on the one hand, the information has been geographically limited to obviate those incidents outside of U.S. jurisdiction, since they are not representative within the whole and could distort the analysis. In this sense, it must be taken into account that they will evaluate the incidents that occurred within the *bounding box* with the following coordinates: Latitude from 15°N to 70°N, Longitude from -180°E to -45°E.

On the other hand, a pre-selection of variables has been carried out to avoid the handling of variables that are not very relevant due to their description or variability. The process followed is shown in a diagram that will be presented at the end of this section in the form of a summary.

All the code used in these tasks is available in the appendices of this report. Source, contents, and pre-selection of data from these three data sources will be explained below.

#### 3.1. USCG: Marine Casualty & Pollution Data for Researchers

The main data for the analysis was obtained from the United States Coast Guard (USCG) website. Specifically, they have a section dedicated to the publication of data for research: Marine Casualty & Pollution Data for Researchers. In this section, they have published two datasets: one up to December 2001 and the other from January 2002 to July 2015. For the study, the latter has been chosen, with the understanding that the closest data will have a greater value when extrapolated to the present. Since the beginning of the 20th century, there have been several updates in the regulations on construction, operation and the environment. To cite some examples, we have Annex IV

of the MARPOL convention that came into force in September 2003, setting the guidelines for the discharge of dirty water or the mandatory double hull for oil tankers.

This dataset contains 10 tables in plain text (.txt format), presented in an 82.5Mb .zip container, which when decompressed occupy a volume of 623 Mb. The R code used for its pre-processing can be found in Annex 3.1. In order to focus this analysis on incident data, the following four tables have been selected:

- **MisleVslEvents**: Contains variables that describe the naval incidents that are the subject of this analysis. 14 variables have been shortlisted:

|                |              |
|----------------|--------------|
| - activity_id  | - waterway   |
| - vessel_id    | - event_type |
| - wine         | - damage_sta |
| - vessel_name  | - latitude   |
| - vessel_class | - longitude  |
| - flag_desc    | - date       |
| - vsl_act_role | - hour       |

- **MisleVessel**: Contains variables relating to the technical and operational characteristics of the vessels involved in the recorded incidents. Regarding the pre-selection of variables, the following 25 variables have been chosen:

|                          |                         |                      |
|--------------------------|-------------------------|----------------------|
| - vessel_id              | - solas_desc            | - build_shipyard     |
| - vessel_name            | - vessel_class          | - build_year         |
| - gross_ton              | - propulsion_type       | - horsepower_ahead   |
| - length                 | - hull_material         | - horsepower_astern  |
| - breadth                | - hull_design_type      | - forebody_type_desc |
| - depth                  | - hull_double_bottom_t  | - hull_configuration |
| - dead_weight_ton        | - hull_double_side_type | - hull_shape         |
| - flag_abbr              | - primary_vin           |                      |
| - classification_society | - imo_number            |                      |

- **MisleVslPoll**: Contains variables about incidents with consequences on the environment (pollution). In this case, the following 14 variables have been pre-selected:

|                |                 |
|----------------|-----------------|
| - activity_id  | - waterway      |
| - vessel_id    | - chris_cd      |
| - wine         | - latitude      |
| - vessel_name  | - longitude     |
| - vessel_class | - discharge_    |
| - flag_desc    | - damage_status |
| - vsl_act_role |                 |

- **MisleInjury**: Contains the variables of incidents with consequences on human beings (injuries and deaths). Finally, the 13 pre-selected variables were the following:

|                |                 |
|----------------|-----------------|
| - activity_id  | - relationship  |
| - vessel_id    | - waterway      |
| - wine         | - accident_type |
| - vessel_name  | - casualty_type |
| - vessel_class | - latitude      |
| - flag_desc    | - longitude     |
| - vsl_act_role |                 |

### 3.2. NOAA: Global Marine data

When evaluating the information contained in the dataset described above, the data regarding the meteorological conditions during each recorded incident were missing. In

the naval environment, it is known that bad weather conditions can cause catastrophes. Three classic examples can be:

- The sinking of the RMS Titanic in 1912: This tragic event is one of the most famous shipwrecks. The Titanic collided with an iceberg in the North Atlantic during a foggy night, leading to the sinking of the luxury liner and the loss of more than 1500 lives.
- The Spanish Armada of 1588 (La Armada Invincible): The Spanish Armada, a fleet sent by King Philip II of Spain with the aim of invading England, was severely affected by a series of storms in the waters around the British Isles. The storms caused the loss of numerous ships and lives, which significantly influenced the failure of the invasion.
- The 1703 English Channel Storm: This storm was one of the most intense to hit southern England. Known as "The Great Tempest," it resulted in the loss of hundreds of ships and thousands of lives.

Because meteorological information is particularly relevant in navigation, it has been decided to include it for this analysis. Specifically, data from marine beacons and sounders managed by the National Oceanic and Atmospheric Administration (NOAA) have been included. These probes collect data such as temperature, wave height, atmospheric pressure, precipitation, wind direction and intensity, etc., either at fixed locations at sea (buoys) or roaming if mounted on ships (mobile stations).

NOAA's publication on this issue consists of .csv files for each station, grouped and compressed monthly into .tar.gz. In this case, the 168 files covering the years from 2002 to 2015 have been considered, with a combined weight of 4.42Gb which, when decompressed, give rise to a total of 79,677 .csv files with a total weight of 56.84Gb in 97,958,906 observations. The code used for the preprocessing of these tables can be found in Annex 3.2.

On this dataset, the following 9 variables have been pre-selected:

- |             |              |
|-------------|--------------|
| - STATION   | - WIND_SPEED |
| - DATE      | - VISIBILITY |
| - LATITUDE  | - AIR_TEMP   |
| - LONGITUDE | - WAVE_HGT   |
| - PAST_WX   |              |

Since the stations record data several times a day and many of the incidents do not have a specific time record, in order to simplify the management of the set, it has been decided to summarize in a single daily record, calculating the average values during the day for continuous variables and the mode for discrete variables.

Later on, it will be verified that while some variables provide relevant information, the statistical analysis will establish that for others this is not so much the case.

### 3.3. NOAA: Daily Summaries

The naval incidents recorded by the USCG cover both those at sea and on waterways (rivers). The Global Marine Data described above does not contain data for rivers. So, to cover this second area, weather data from ground stations through NOAA's Daily Summaries publication have been included.

Specifically, the river areas to be explored are: Mississippi and Ohio. These areas have different characteristics from those found in the sea. For example, the sandbars of the

Mississippi River are known to cause ships to run aground or sink. In addition, the absence of waves and salinity influence the construction requirements of the ships. The materials used, design and hull size of boats destined for rivers are, in general, different from those destined to operate at sea. For this reason, marine and river incidents will be analysed separately.

The Daily Summaries data provided by NOAA's ground stations include data similar to the above, except for wave height, which, in this case, would not make sense.

The format presented by NOAA is .csv.gz files of annual sets. 14 files (from 2002 to 2015) have been collected, with a total weight of 2.17Gb. The code used for its pre-processing is available in Annex 3.3. When decompressed, 18.76Gb of data were obtained: 154,188,126 observations with these 8 pre-selected variables:

- |           |             |
|-----------|-------------|
| - STATION | - PRCP      |
| - DATE    | - AWND      |
| - TMAX    | - LONGITUDE |
| - TMIN    | - LATITUDE  |

The geographical boundary for Mississippi has been: Latitude from 30.5°N to 45°N, Longitude from -88°E to -94°E. For Ohio, the following values correspond: Latitude from 37.5°N to 41°N and Longitude from -79.5°E to -88°E, has been reduced to 12,281,982 observations.

### 3.4. Data merging: obtaining the datasets

To obtain the two datasets that serve as the basis for the analysis, the data from the three sources described above have been combined, using the `MisleVslEvents` table as a basis. The R code used for this work is available in Annex 3.4. At this point, it would be advisable to keep in mind the preprocessing diagram presented at the end of this section to see the relationships between tables, as well as the discarding of variables carried out in each step.

In order to facilitate data processing in a more agile way, an intermediate table "Events" has been created that collects filtered data from events without repeating and excluding those events that have not occurred in the main areas of this analysis: oceanic areas of the USA and the Mississippi River basin. The *rnaturalearth* library has been used to delimit these areas to match the meteorological data.

Both naval incidents and meteorological data are geolocated by coordinates. These coordinates do not coincide exactly, so in order to assign weather conditions to each incident, the distance to the nearest station has been calculated using the following approximation:

$$Distancia = \sqrt{(Lat_1 - Lat_2)^2 + (Long_1 - Long_2)^2}$$

Once the geographical dimensions, the pre-selection of variables and the vertical aggregation of the observations have been carried out, the two sets of data mentioned above have been obtained:

1. **VesselBalancedSample:** There are 109,836 observations with data from 54,198 ships involved in incidents and another 54,198 ships with no recorded incidents. The data for the 54,198 ships involved in incidents comes from unique `vessel_id` records in the Events table. On the Vessel board, an exclusive difference has been made to obtain the `vessel_id` that are not common between both boards,

understanding that boats that do appear in Vessel but do not appear in Events, are exempt from incidents. In order not to bias the sample, a random selection of non-incident ships of the same size as the total number of those involved in incidents has been made. The result has been a sample with 50% of ships with incident + 50% of ships without incident. It contains 12 variables that, on this occasion, will be described and subjected to a descriptive statistical analysis in the following section of this document:

- vessel\_id: Unitary identification of the vessel, assigned by the USCG. It is used for the purposes of this database, and this data is unique to each ship. This data is not used outside this scope, so it is useful for analysis, but not for exporting information from it.

- imo\_number: Unitary identification of the ship, assigned to the IMO (International Maritime Organization). This identification is universal and is frequently used by various organizations and companies to identify the name. It is worth mentioning that this data does not vary during the life of the boat, but not all boats are obliged to have it. This causes absences in the database relatively frequently. In general, ships required to have an IMO (International Maritime Organization) number are those that meet one of the following criteria:

- Merchant ships of 300 gross registered tons (GT) or more.
- Passenger vessels of 100 GT or more.
- Cargo ships of more than 500 GT carrying hazardous substances or noxious materials.
- Mobile offshore drilling rigs.

Warships, fishing vessels and small pleasure craft are generally not required to carry the IMO number.

- vessel\_name: Name of the ship. This information facilitates a quick identification, but as it can be changed during the life of the same, it is not a better building criterion. Names that are very similar to each other may even appear for convenience, so this is for informational purposes only.

· School Ship

- |                     |                      |                 |
|---------------------|----------------------|-----------------|
| · Barge             | · Passenger          | · School Ship   |
| · Bulk Carrier      | · Recreational       | · Tank Ship     |
| · Fishing Vessel    | · Refrigerated Cargo | · Towing Vessel |
| · General Dry Cargo | · Research Ship      | · UNSPECIFIED   |
| · Miscellaneous     | · Ro-Ro Cargo Ship   | · Warship       |
| · Offshore          | · Passenger          | · School Ship   |

- build\_year: Year of construction. With this variable, the age at the time of the incident can be calculated.

- gross\_ton: Gross Tonnage (GT) measures the total volume of a ship, including internal spaces. It is a metric to assess your size, but it also determines other parameters such as applicable legislation, crew size, etc.

- length: Length or length of the vessel measured in feet. Another size measurement that describes the vessel. We have chosen to omit the beam (or width) since there is a directly proportional relationship between both magnitudes.

- flag\_abbr: Flag Abbreviation. It is the flag under which it operates. The flag determines the applicable laws, regulations, and protections to each ship.

Therefore, it is possible to evaluate how more or less restrictive legislation affects the health of each vessel.

- **class\_soc**: Classification Society. Classification societies provide certification services to ensure that ships meet safety and quality standards. These certificates are crucial for global acceptance and safe operation of vessels.

- **solas\_desc**: Accession to SOLAS, the International Convention for the Safety of Life at Sea, which establishes maritime safety standards to prevent accidents and safeguard lives on international ships. It is understood that, if a ship is under this convention, its safety requirements meet universally accepted standards.

· **Damage to Cargo**

- |                     |                        |                     |
|---------------------|------------------------|---------------------|
| · Grounding         | · Sinking              | · Loss of Stability |
| · Environment       | · Emergency            | · Damage to Cargo   |
| · Failure (Vessels) | · Loss of Power        | · Explosion         |
| · Allision          | · Evasive Maneuvers    | · Failure (Diving)  |
| · Set Adrift        | · Failure (Nonvessels) | · Blowout           |
| · Maneuverability   | · Fouling              | · Falls into Water  |
| · Collision         | · Abandonment          | · Implosion         |
| · Flooding          | · Capsize              | · Loss of Stability |
| · Fire              | · Sinking              | · Damage to Cargo   |

- **damage\_status**: Indicates whether the boat has been damaged or not. Incidents in which a ship is involved are frequent and do not result in damage itself, so it is considered relevant that this aspect is evaluated.

2. **MergedActivity**: These are 68,000 observations with detailed information on the recorded incidents. This table condenses the ships' characteristics data along with incident descriptions and weather information. In this sense, the best option would have been to have detailed information on "non-incident" events to establish a "non-incident" level. However, from the data of ships without incident, there are no accessible position and weather records. At a certain stage of this analysis, it was proposed to generate synthetic data, but it was discarded because these data would correspond to 100% of an entire class of the predictive variable, preventing the objective of transfer and real application of the analysis. This dataset contains 28 variables, which, like the previous table, will be described and statistically analyzed later:

- **activity\_id**: Identifier issued by the USCG for each recorded incident. This variable is included as a link between the **vessel\_id** involved in the same incident.

- **date**: Date of the incident. The date of the year determines the weather and commercial activity. In this case, it has been simplified to the month when the event occurred to reduce variability.

- **hour**: Time of incident. As in the previous case, the time determines more or less exposure to activity. To reduce variability, minutes have been dispensed with.

· **West Coast**

- |          |                  |               |
|----------|------------------|---------------|
| · Alaska | · East Coast     | · Mississippi |
| · Canadá | · Gulf of Mexico | · West Coast  |

- Longitude: Coordinate of longitude, measured in degrees.
- latitude: Latitude coordinate, measured in degrees.
- watertype: Type of aquatic environment: River or Ocean. Variable calculated thanks to the rnatrualearth library. To do this, data have been obtained from the polygon formed by the USA and its neighbouring countries. The points of longitude and latitude within the polygon have been considered river; Ocean is everyone else.
- event\_type: Type of incident recorded. In this case, since there are no uneventful observations, there is no "No event" value.
- damage\_status: *(ver descripción anterior)*
- vessel\_id: *(ver descripción anterior)*
- imo\_numer: *(see description above)*
- vessel\_name: *(ver descripción anterior)*
- vessel\_class: *(ver descripción anterior)*
- build\_year: *(ver descripción anterior)*
- age: Age of the vessel at the time of the incident, calculated as the date of the incident minus the date of construction (build\_year).
- gross\_ton: *(ver descripción anterior)*
- Lenght: *(see description above)*
- flag\_abbr: *(ver descripción anterior)*
- classif\_society: *(ver descripción anterior)*
- solas\_desc: *(see description above)*
- air\_temp: Ambient temperature recorded during the incident, measured in degrees Fahrenheit (°F). Low temperatures give an idea of more adverse conditions.
- wind\_speed: Wind speed, measured in knots. Likewise, high values of this variable would imply worse conditions at sea, but also in rivers.
- wave\_hgt: Wave height, measured in feet. This value has only been collected for incidents at sea, as it is understood not to be significant in rivers.
- visibility: Visibility according to the WMO (World Meteorological Organization) scale. With a scale of 90 to 99, it measures the distance that is visible to the naked eye, with 90 being the equivalent of 0.05km and with 99, it would be up to more than 50km.
- damage\_assessment: Economic dollar value of the damages caused by each accident. The information is collected by the commissions of inquiry on the basis of data provided by the insurance companies involved.
- casualty: Description of personal injury. Mainly this value can be identified with death, injury or loss.
- pollution: A type of substance spilled into water.
- event\_class: Incident type produced: Critical Events / Onboard Emergencies / Maritime Accidents / Material Issues / Third-party Damages. It is a convergence

of the event\_type variable as it contains up to 26 levels that were hindering data processing. To achieve this convergence, the following conceptual rule has been established:

1. Critical Events:

Sinking, Implosion, Capsize, Loss of Stability, Vessel Maneuverability, Set Adrift, Abandonment.

2. Onboard Emergencies:

Loss of Electrical Power, Fire, Emergency Response, Explosion, Flooding, Personnel Casualties, Falls into Water.

3. Maritime Accidents:

Grounding, Allision, Collision.

4. Material Issues:

Material Failure Vessels, Material Failure Non-vessels, Material Failure Diving, Blowout.

5. Third-party Damages:

Damage to the Environment, Damage to Cargo, Fouling, Evasive Maneuvers, UNSPECIFIED

Below, as a summary of this section, a diagram is presented showing the characteristics and dependencies of the tables used to arrive at the two sets of data on which the analysis is based:

## Preprocessing Diagram

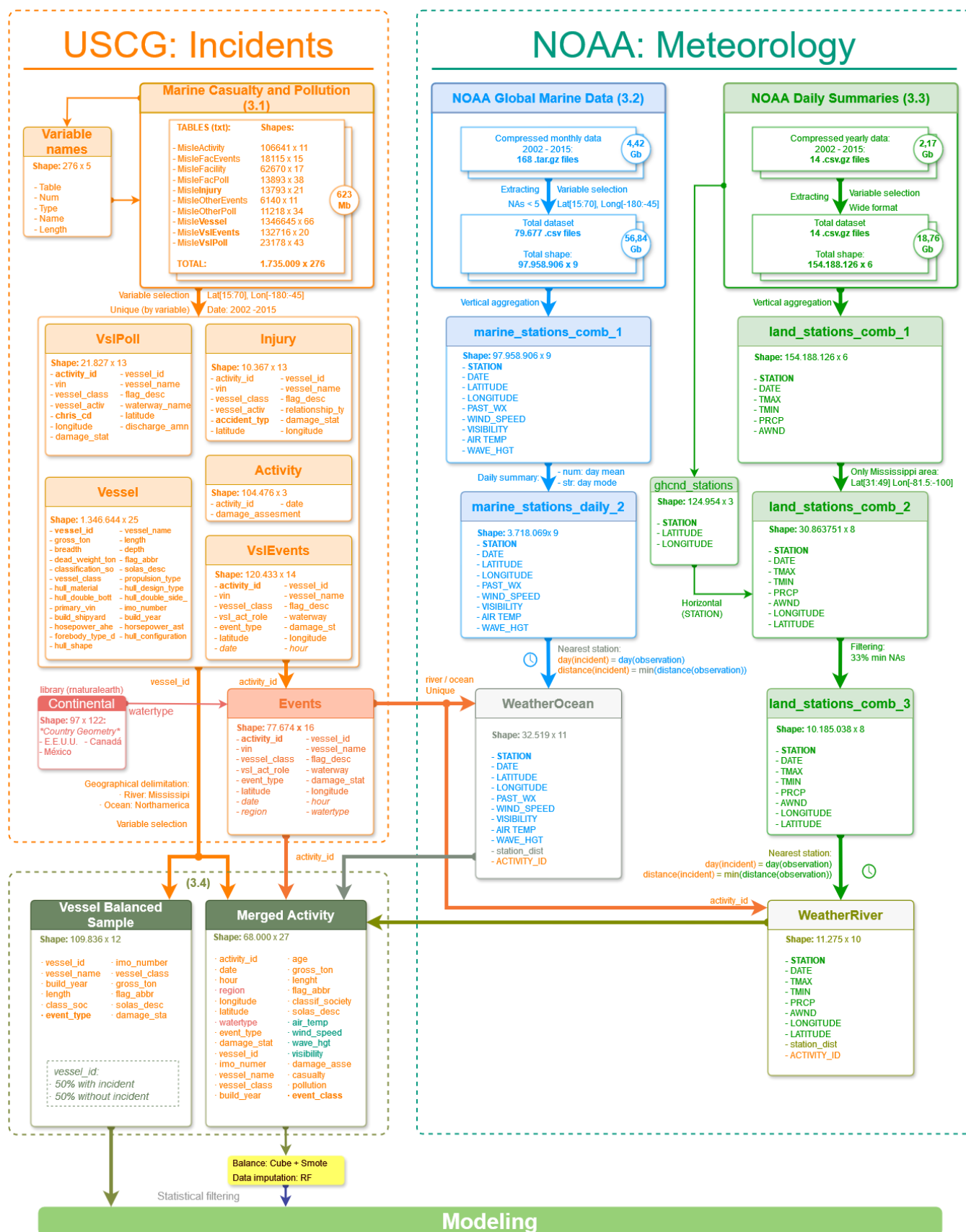


Fig.3.4.:Preprocessed diagram

## 4. Exploratory Data Analysis (EDA)

In order to make as accurate a portrait as possible of the data analysed, different statistical aspects of the datasets will be presented. These aspects will be instrumentalized through graphs and tables obtained with Python, using libraries such as matplotlib. The code used is available in the final appendices of this report.

### 4.1. Dataset VesselBalancedSample

This dataset consists of 109,836 observations with data from 54,198 ships involved in incidents and another 54,198 ships with no recorded incidents. With this data, the aim is to establish a model that predicts whether a given ship will be involved in an incident. The most important results of the statistical exploration of its variables will be presented below, but the complete code can be found in Annex 4.1.

#### Statistical summary (ydata\_profiling report)

| Dataset statistics                                                                     |          | Variable types   |   |
|----------------------------------------------------------------------------------------|----------|------------------|---|
| Number of variables                                                                    | 13       | Numeric          | 5 |
| Number of observations                                                                 | 109836   | Text             | 3 |
| Missing cells                                                                          | 0        | Categorical      | 5 |
| Missing cells (%)                                                                      | 0.0%     |                  |   |
| Duplicate rows                                                                         | 0        |                  |   |
| Duplicate rows (%)                                                                     | 0.0%     |                  |   |
| Total size in memory                                                                   | 10.1 MiB |                  |   |
| Average record size in memory                                                          | 96.0 B   |                  |   |
| Alerts                                                                                 |          |                  |   |
| build_year is highly overall correlated with vessel_id                                 |          | High correlation |   |
| classification_society is highly overall correlated with solas_desc                    |          | High correlation |   |
| gross_ton is highly overall correlated with length                                     |          | High correlation |   |
| length is highly overall correlated with gross_ton and 1 other fields                  |          | High correlation |   |
| solas_desc is highly overall correlated with classification_society and 2 other fields |          | High correlation |   |
| vessel_class is highly overall correlated with solas_desc                              |          | High correlation |   |
| vessel_id is highly overall correlated with build_year                                 |          | High correlation |   |
| classification_society is highly imbalanced (87.7%)                                    |          | Imbalance        |   |
| solas_desc is highly imbalanced (57.7%)                                                |          | Imbalance        |   |
| damage_status is highly imbalanced (59.0%)                                             |          | Imbalance        |   |
| df_index is uniformly distributed                                                      |          | Uniform          |   |
| df_index has unique values                                                             |          | Unique           |   |

Fig. 4.1. VBS Summary

One of the noteworthy facts is that, in this case, all the variables have complete data, which is especially favorable for the application of machine learning algorithms.

#### 4.1.1. Characteristics of the vessels

##### Class of vessel (vessel\_class)

What types of boat have been considered in the analysis? Below is a bar graph with type of vessel, ordered from highest to lowest frequency and distinguishing those vessels that have been the subject of incidents:

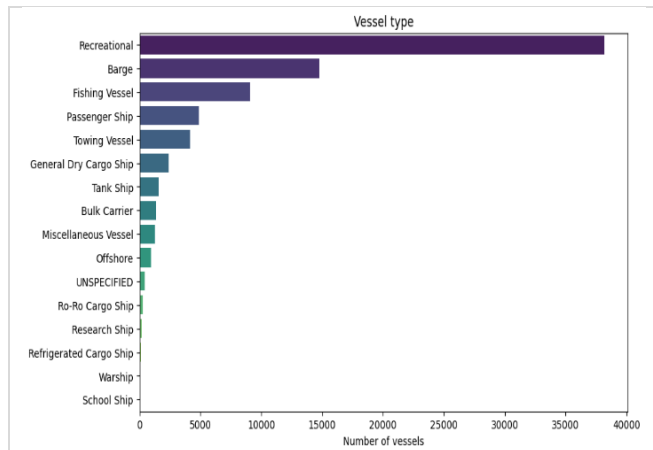


Fig. 4.1.1.1.: VBS vessel\_class

*The most frequent type of boat is the recreational one, with almost 40,000 boats registered. However, they are the least affected by incidents*

##### Year of construction (build\_year)

How old are these boats? It can be verified with the following bar graph in which the involvement in accidents has been distinguished

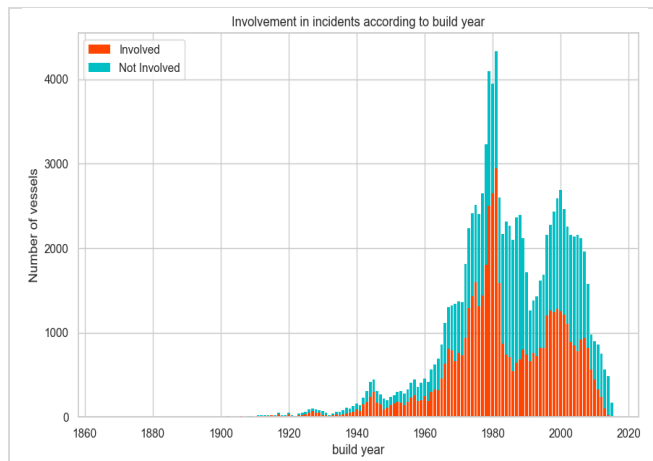


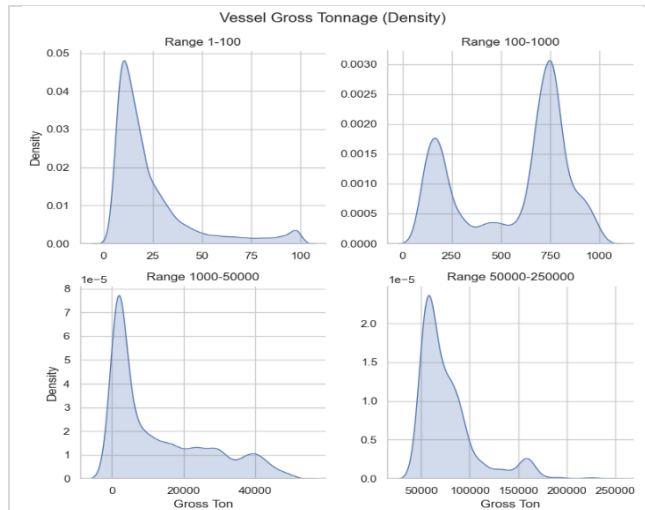
Fig. 4.1.1.2.: VBS build\_year

*In this case, it can be seen that the majority of vessels are concentrated around the beginning of the 80s and after a drop, they grow again around the year 2000.*

*The distribution of those affected by incidents / not affected has a similar trend*

## Size: Gross Tonnage and length

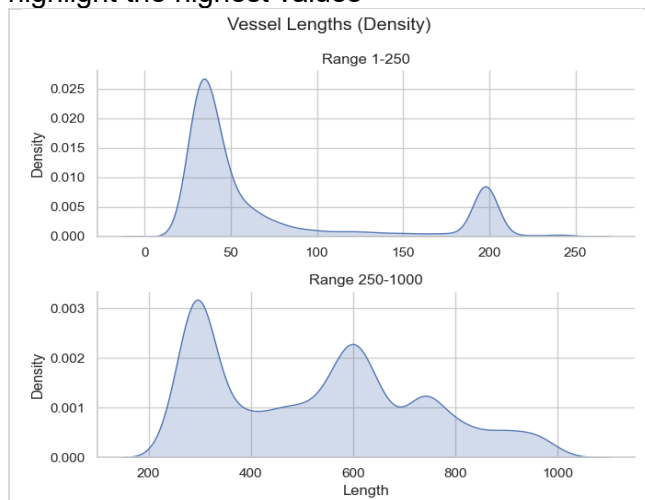
As for the dimensions, on the one hand, we are going to see the Gross Tonnage (or Gross Tonnage), which refers to the volume of the boat, divided into four sections since there are many small boats that prevent the complete image from being correctly assessed.



Here the majority are concentrated in the 0 and 25 GT, with a notable peak at 750GT

Fig. 4.1.1.3.: VBS Gross Tonnage

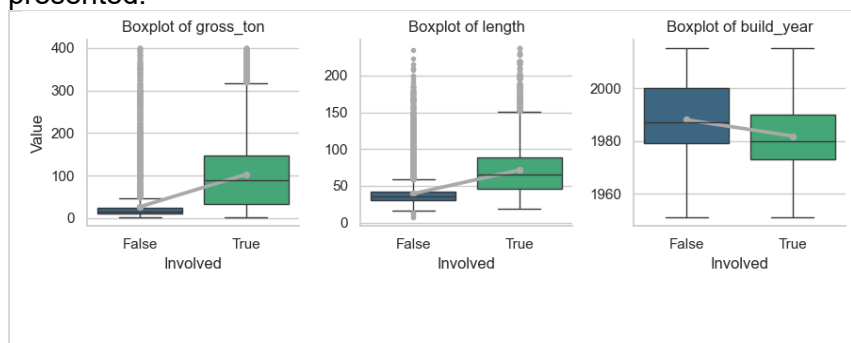
Regarding the length, we are going to present two density graphs with two sections to highlight the highest values



It can be seen how the majority of boats are concentrated around 30-40 feet in length.

Fig. 4.1.1.4.: VBS Length

To evaluate the most frequent values of the previous variables, some box plots will be presented:



In these variables, differences in means are seen depending on whether the vessel is involved in accidents, which is why they were included in the analysis.

Fig. 4.1.1.5.: VBS Box-Plots Vessel

### 4.1.2. Legal and regulatory aspects

#### Flag (flag\_abbr)

As for the flag, a graph is presented that counts ships under each of the most frequent flags:

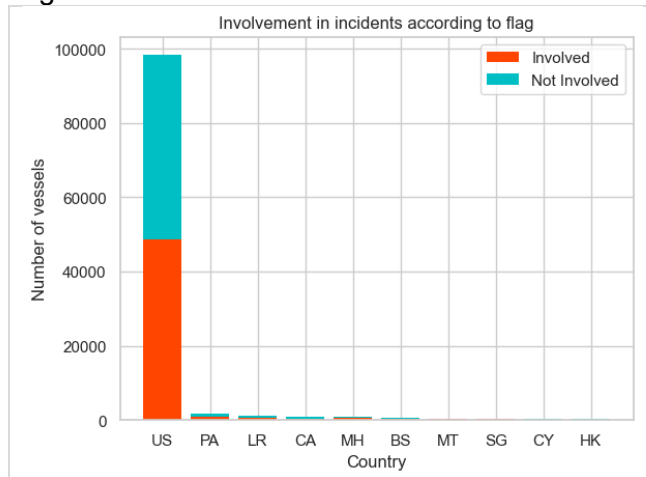


Fig. 4.1.2.1: VBS Flag

When the analysis focuses on the United States, the vast majority of flags are local due to legislative issues, but in the rest of the world this situation does not occur. In any case, other notable flags are:

- PA - Panama
- CA - Canada
- LR - Liberia

No differences are seen for involvement in incidents.

#### Classification Society

In this case, distribution percentages have also been added, since differences can be seen for the impact on incidents.

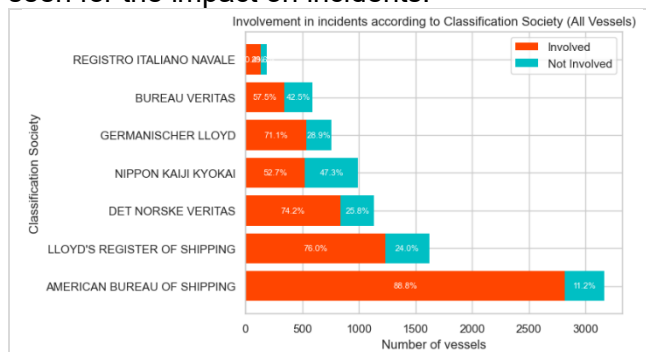


Fig. 4.1.2.2.: VBS Clasification

In this case, ABS takes the most classifications. However, differences can be seen for involvement in incidents

#### SOLAS membership (solas\_desc)

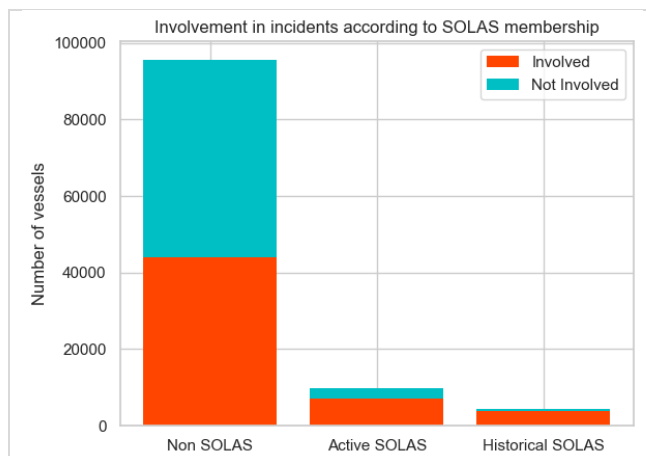


Fig. 4.1.2.3.: VBS Solas

Here you can see how there is a majority that is not required to comply with SOLAS. However, for those who are (Active Solas), there is a paradoxical difference between incident involvement.

### 4.1.3. Characteristics of incidents

#### Incident type (event\_type)

What are the most common types of accidents? In the following bar graph you can see the incidents, ordered from highest to lowest frequency and distinguishing those vessels that have been the subject of incidents:

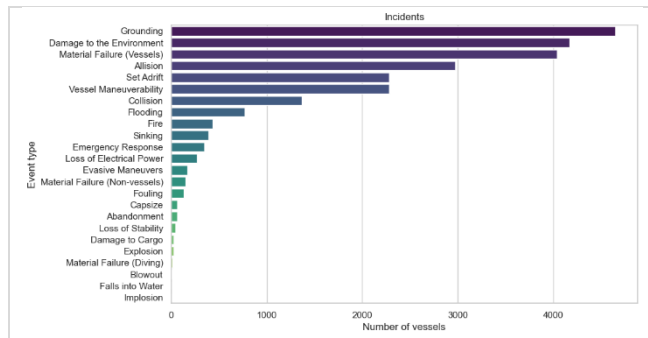


Fig. 4.1.3.1.: VBS Incidents

*The majority of incidents correspond to Grounding, followed by Damage to the Environment and Material Failure.*

#### Damages (damage\_status)

As seen in the previous graph, a large part of these incidents are environmental damage. There may be cases in which these events occur without causing material damage to the boat. For example, if certain valves are opened by mistake, there may be a harmful spill, but it does not cause damage. Let's look at this data in more detail:

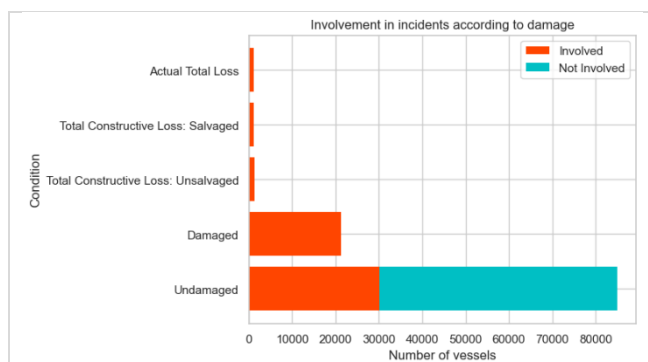


Fig. 4.1.3.2.: VBS Damages

*In this case, it can be seen how 35.4% of the ships involved in accidents have not recorded damage. On the other hand, logically, the ships with recorded damage have all been involved in incidents.*

Note: Other variables such as ship name or IMO number are included in this dataset that have no value for this analysis, but are retained as information for possible data extrapolations.

#### 4.1.4. Correlations

Correctly understanding the correlations in a dataset is critical before applying machine learning models. This step is essential for identifying patterns, relationships, and dependencies between variables, providing crucial information for the effective design of predictive models. Analyzing correlations allows you to detect possible biases, avoid multicollinearity, and improve the interpretation of results. With this process, informed decision-making and optimization of model performance is ensured.

In this case, the PerformanceAnalytics library has been used, which provides additional data to the correlations such as correlation significance and histograms. The result is the following table:

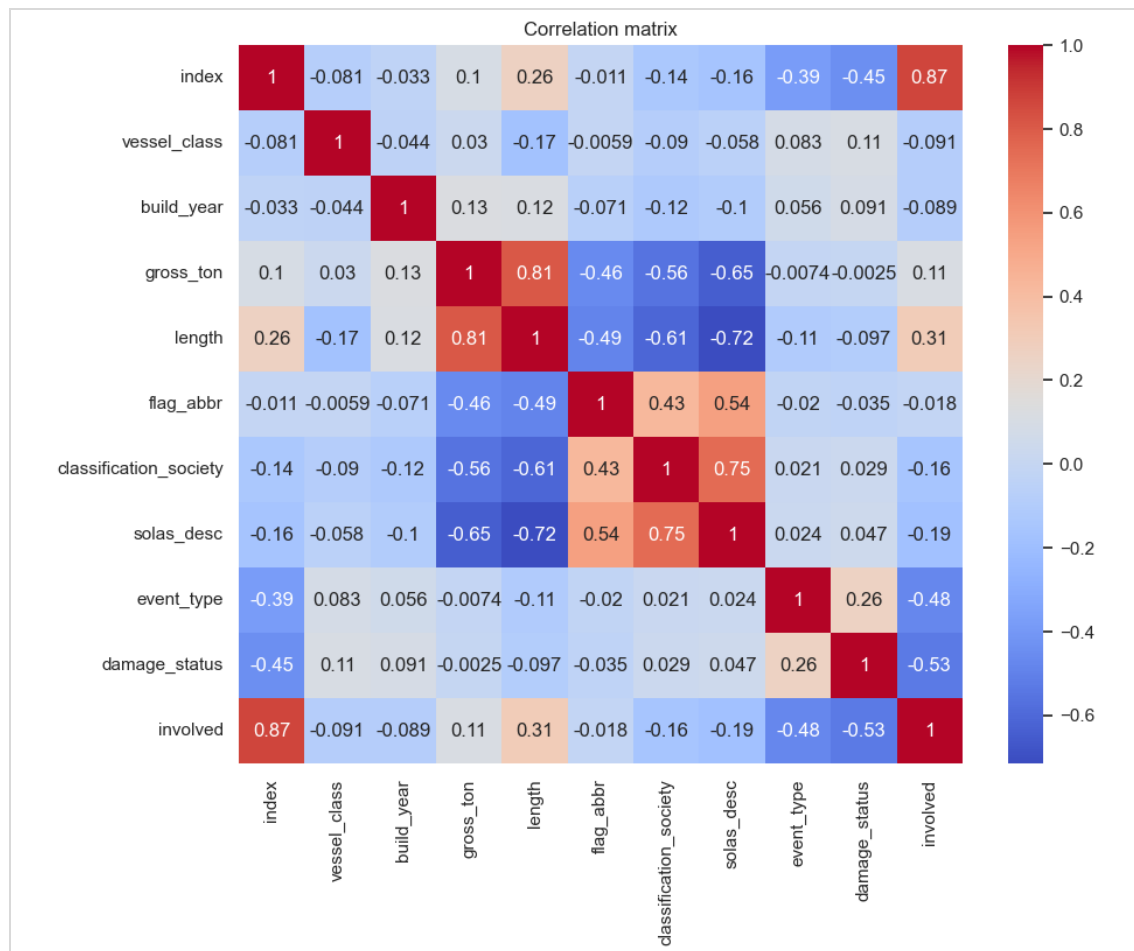


Fig. 4.1.4. VBS Correlations

There are high and significant correlations between:

- length and gross\_ton: As expected, a greater length implies a greater volume, since these magnitudes are proportional to the design itself. In any case, we are going to choose to keep both as they do not include beam, draught or freeboard, leaving certain degrees of freedom for the physical characteristics of the boats.
- classification\_Society and solas\_desc: Normally, ships with higher volumes are required to abide by both issues, so this ratio is within what is expected. Similarly, for analysis, they will be retained to assess whether the different standards of classification societies have an influence on improving safety at sea.

## 4.2. Dataset MergedActivity

This other dataset is made up of a total of 68,000 observations and 26 variables that describe the incidents in detail. By focusing on the description of incidents, there is no data for ships without incidents. With this data, the aim is to predict what type of incident will occur when certain circumstances or characteristics are met. It has several variables in common with the previous dataset, so its descriptions will not be repeated. However, all of them have been analysed and the code used is available in Annex 4.2.

### Statistical summary (ydata\_profiling report)

| Dataset statistics            |          | Variable types |    |
|-------------------------------|----------|----------------|----|
| Number of variables           | 29       | Numeric        | 13 |
| Number of observations        | 68000    | Date Time      | 3  |
| Missing cells                 | 269764   | Categorical    | 9  |
| Missing cells (%)             | 13.7%    | Text           | 4  |
| Duplicate rows                | 0        |                |    |
| Duplicate rows (%)            | 0.0%     |                |    |
| Total size in memory          | 14.0 MiB |                |    |
| Average record size in memory | 216.0 B  |                |    |

### Alerts

|                                                                                        |                  |
|----------------------------------------------------------------------------------------|------------------|
| latitude is highly overall correlated with region and 1 other fields                   | High correlation |
| length is highly overall correlated with gross_ton and 2 other fields                  | High correlation |
| longitude is highly overall correlated with region and 1 other fields                  | High correlation |
| region is highly overall correlated with latitude and 2 other fields                   | High correlation |
| solas_desc is highly overall correlated with classification_society and 2 other fields | High correlation |
| vessel_class is highly overall correlated with solas_desc and 1 other fields           | High correlation |
| classification_society is highly imbalanced (82.7%)                                    | Imbalance        |
| casualty is highly imbalanced (50.1%)                                                  | Imbalance        |
| air_temp has 5342 (7.9%) missing values                                                | Missing          |
| wind_speed has 25494 (37.5%) missing values                                            | Missing          |
| wave_hgt has 55164 (81.1%) missing values                                              | Missing          |
| visibility has 63031 (92.7%) missing values                                            | Missing          |
| casualty has 65628 (96.5%) missing values                                              | Missing          |
| pollution has 55049 (81.0%) missing values                                             | Missing          |
| damage_assessment is highly skewed ( $\gamma_1 = 83.74039342$ )                        | Skewed           |
| age has 834 (1.2%) zeros                                                               | Zeros            |
| damage_assessment has 38580 (56.7%) zeros                                              | Zeros            |

#### 4.2.1. Location

One of the most characteristic pieces of information about incidents is where they occurred. To do this, as in this case, geoposition data with latitude and longitude are usually used. These numerical values provide a single reference for any point on the Earth's surface. Latitude measures the distance north or south of the equator, while longitude indicates the position east or west of the Greenwich meridian. Together, these coordinates form a universal geographic code that allows us to navigate the world accurately and locate any place accurately, facilitating mapping, navigation, and communication.

Regarding the screening of observations, it should be noted that, during the statistical exploration, some incidents were detected recorded in very remote and even unlikely places such as in the middle of a plain. With the help of the `rnaterearth` library, it has been possible to differentiate the points corresponding to the continental and maritime zone. Thanks to this data cleansing, only the regions of: Alaska, Canada, East Coast, Gulf of Mexico, Mississippi (only river area) and West Coast will be considered.

For the representation of maps, the `leaflet` library has been used, which offers the possibility of creating interactive maps. As this is a static document, captures of the maps will be used, thus losing the details in the form of a pop-up for each point.

#### Regions

The characteristics of the aquatic environment are very different in the areas of Alaska from those of the Gulf of Mexico or the Mississippi River. For this reason, it was decided to create a variable that encompasses similar marine environments. In this case, the following have been determined:

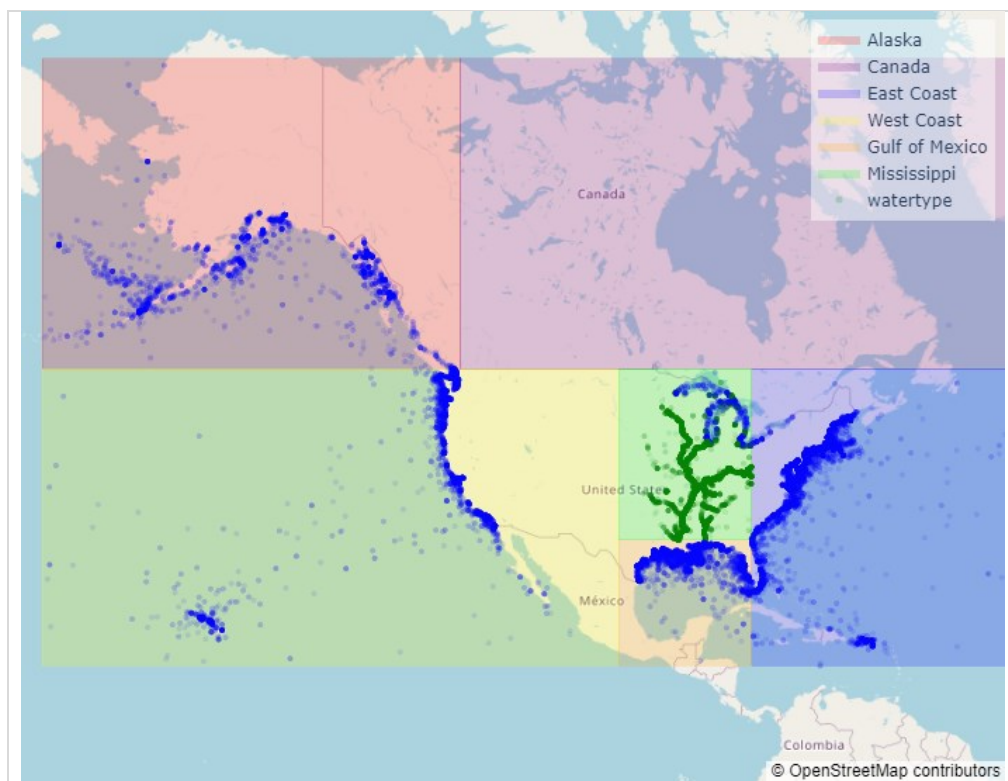


Fig. 4.2.1.1 MA Regions

Regarding the distribution of the number of accidents, due to the overlap of points and the random sample used to lighten the load, it is not possible to determine the number of incidents on the map. However, it can be verified using the following bar graph:

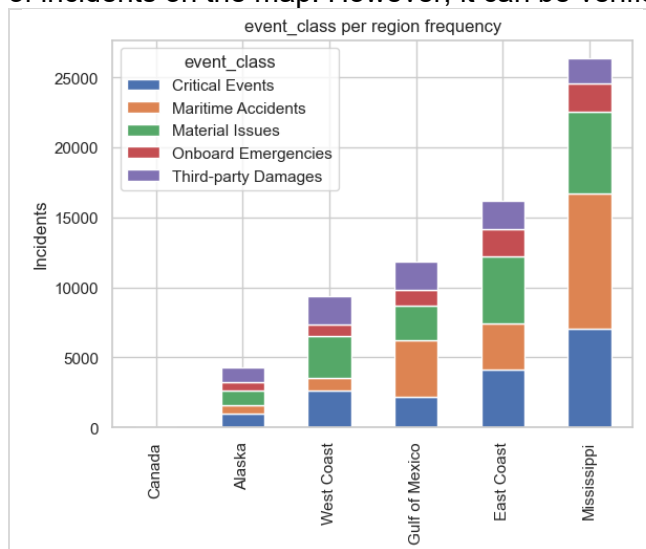


Fig. 4.2.1.2.: MA Regions incident

The area with the most recorded incidents is Mississippi with more than 25,000. Regarding the distribution of the type of incidents, it can be highlighted that data to third parties maintains different proportions depending on the regions.

Same information over a map:

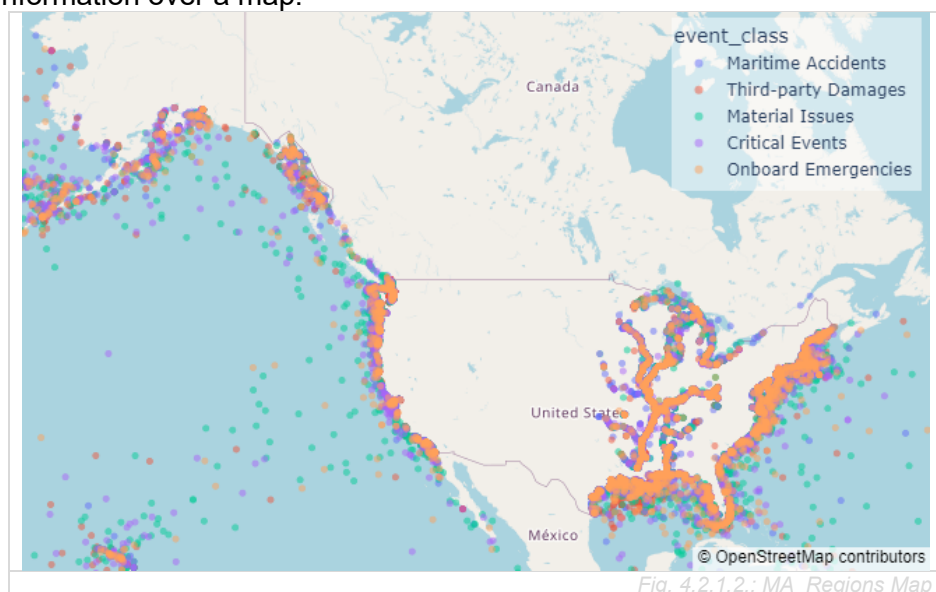


Fig. 4.2.1.2.: MA Regions Map

### Most common event in each region (event\_type)

What is the most common incident in each region? Most frequent events:

| region         | most_frequent_event        | num_events_per_region |
|----------------|----------------------------|-----------------------|
| Mississippi    | Grounding                  | 11374                 |
| East Coast     | Material Failure (Vessels) | 9630                  |
| Gulf of Mexico | Damage to the Environment  | 7163                  |
| West Coast     | Material Failure (Vessels) | 5900                  |
| Alaska         | Damage to the Environment  | 2692                  |
| Canada         | Material Failure (Vessels) | 1                     |

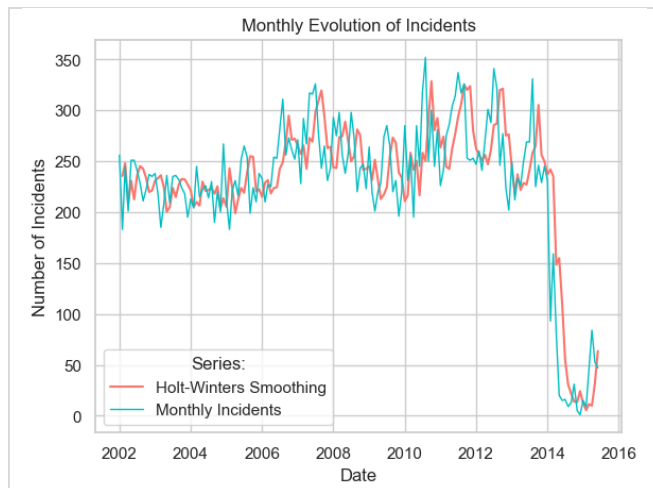
Fig. 4.2.1.3.: MA Most frequent Incidents

Groundings are common in Mississippi area due to its shallow draft

### 4.2.2. Chronology

#### Time series (date)

In this data set, there is date and time information for each incident, so the temporal evolution can be evaluated through the following series to which a Holt-Winters type smoothing was added. Holt-Winters smoothing is a forecasting method for time series that considers trend and seasonality. Uses weighted averages and adjusts smoothing factors to accurately predict future values. In this case, the smoothing is not fully adjusted, so some randomness can be inferred.

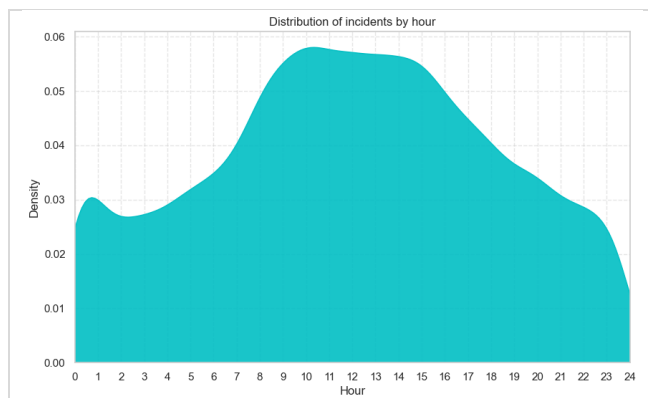


*The most notable thing is a notable drop in incidents starting in 2014.*

Fig. 4.2.2.1.: MA Time series

#### Hourly Profile

Regarding the time at which incidents occur, knowing when they are most frequent can help take measures to avoid them. The following graph shows the time profile that they follow

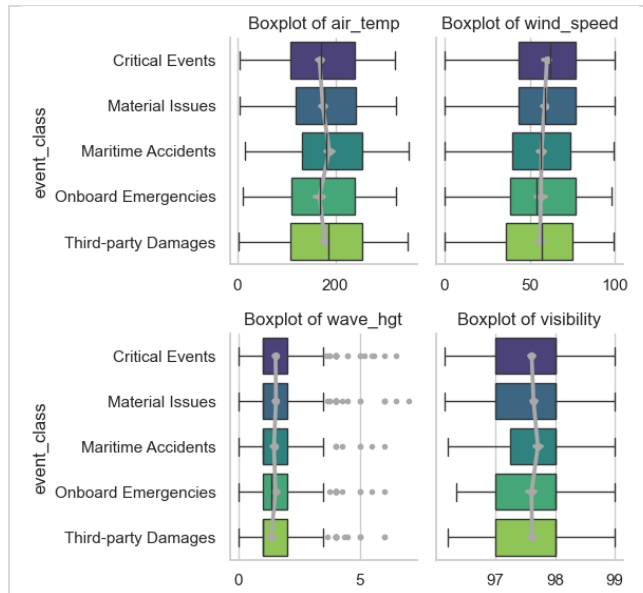


*As suspected, incidents are more frequent in the middle of the day*

Fig. 4.2.2.2.: MA Daytime profile

### 4.2.3. Meteorology

After processing and joining meteorological data obtained from NOAA, these can be included for analysis, enriching the original information. Following graph shows their most frequent values, according to type of event:

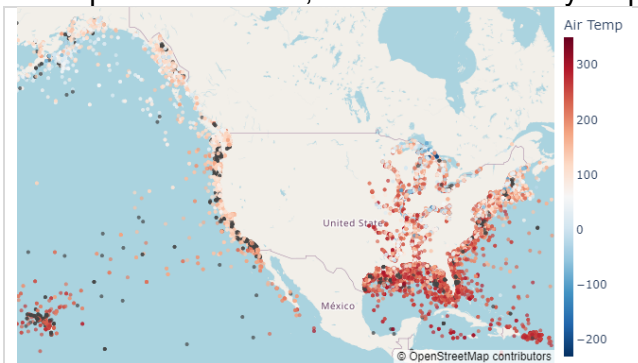


*It seems that the visibility and hour height variables do not have significantly different values depending on the type of incident, which is why they are candidates to be excluded in the machine learning models.*

Fig. 4.2.3.1.: MA Box-Plot Meteorology

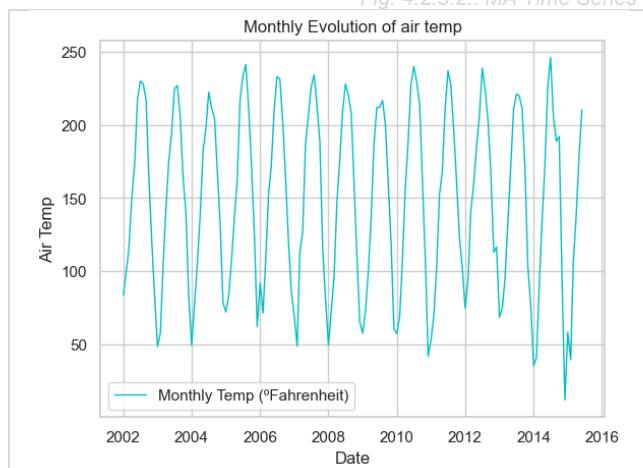
### Temperature (air\_temp)

If you consider the temperature separately, looking at its geographical distribution and its temporal distribution, it does not offer any surprises:



*As expected, the lowest temperatures are located at a higher latitude, in Canada, and the highest temperatures are located further south in the Gulf of Mexico.*

Fig. 4.2.3.2.: MA Time Series



*The average monthly temperature follows annual cycles that are very similar to each other, so estimates of this parameter can be made.*

Fig. 4.2.3.3.: MA Temp series

#### 4.2.4. Characteristics of the vessels

Although most of its variables have been reviewed in section 4.1, as new variables are included in this dataset, they can be evaluated with these in mind.

##### Age (date – build\_year)

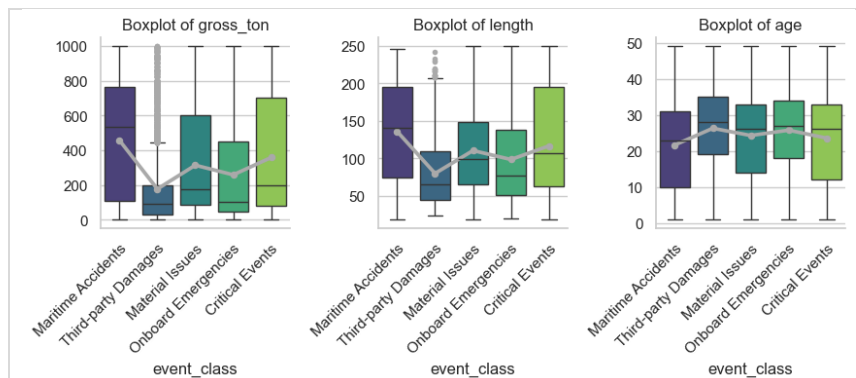
For example, with the construction date and the incident date, you can obtain the age



*Two frequency peaks can be seen: very new boats and boats that are around 30 years old.*

Fig. 4.2.4.1.: MA Age

Let's see how they are distributed according to the type of incident recorded, along with the other two design characteristic variables: gross\_ton (volume) and length (length)

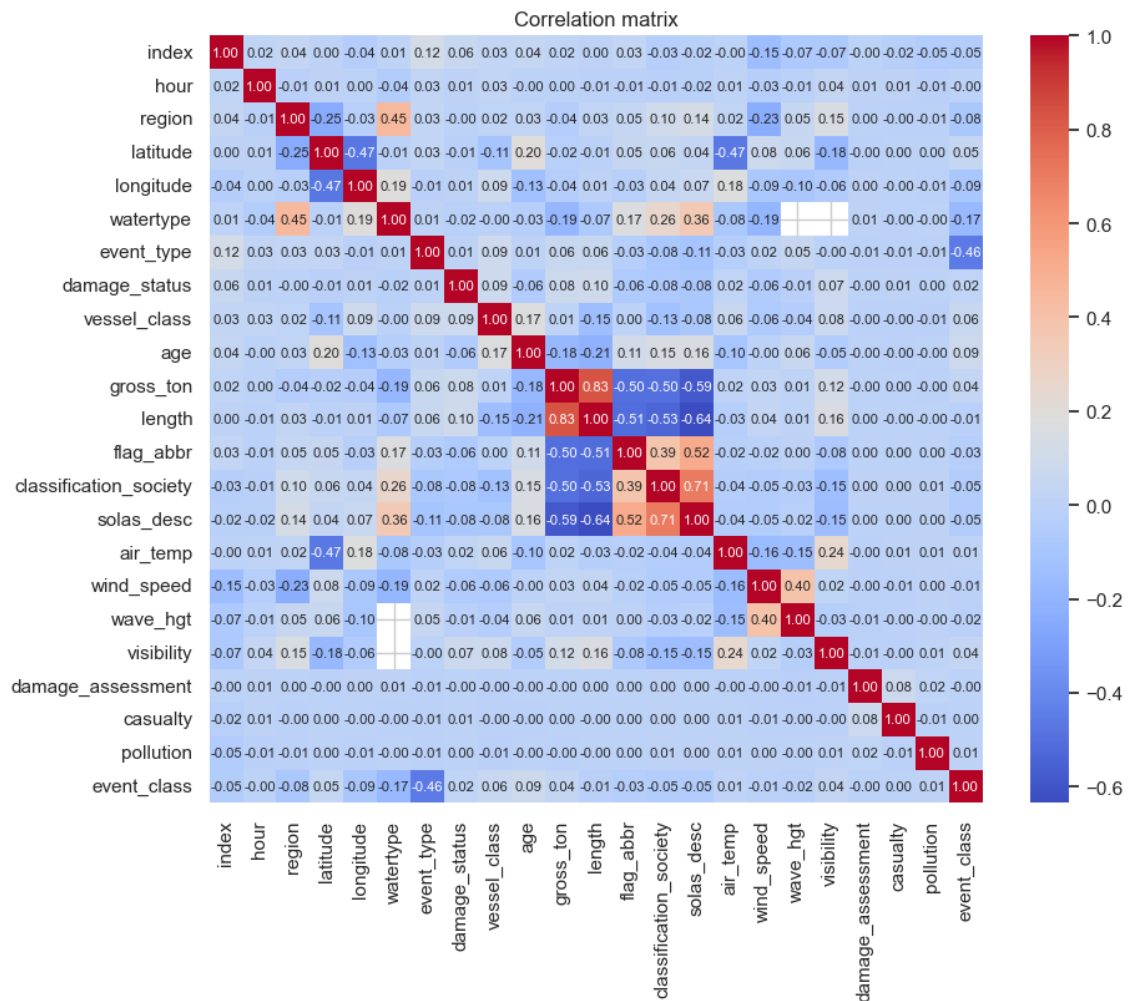


*In the three variables there are differences in their most frequent values, so they will not be discarded for processing.*

Fig. 4.2.4.2.: MA Box-Plots Vessels

#### 4.2.5. Correlations

As established above, the relationship of the variables is a fundamental aspect for the application of the algorithms that will be discussed in the following sections.



For this dataset, the relationships between the ship characteristics variables exhibit the same patterns as in the previous set. In terms of meteorological variables, there is a certain correlation between wave height (wave\_hgt) and wind (wind\_speed), as is usual around the world.

## 5. Application of Machine Learning Models

---

In this section, the results of the application of machine learning algorithms will be presented, as well as a performance comparison based on the most common metrics in this field.

Machine learning models are algorithms and computer systems designed to learn patterns and perform specific tasks without being explicitly programmed. Instead of relying on static programming rules, these models use data to learn and improve their performance over time.

These models have applications in a wide variety of fields, from speech recognition, image recognition, personalized recommendations, autonomous vehicles, or predictive analytics as in the case of this analysis. Machine learning is transforming the way we interact with technology and how we approach complex problems across disciplines.

The learning process involves feeding the model a set of input data, allowing it to identify patterns and make predictions or decisions. There are a variety of machine learning approaches, such as supervised learning, where the model is trained on labeled examples; unsupervised learning, where the model finds patterns without labels; and reinforcement learning, where the model makes decisions based on the feedback received.

In the process of developing machine learning models, a training dataset and a test dataset are used to evaluate and validate the model's performance. The training dataset is used to teach the model patterns and relationships between variables, allowing it to learn from the behavior of the data. Once trained, the model is evaluated using the test dataset, which has not been viewed during the training phase. This independent set is used to test the model's ability to generalize and make accurate predictions on unfamiliar data. The split between training and testing helps identify potential overfitting issues, where the model adapts too much to the training data, but doesn't generalize well to new data. This cross-validation approach improves confidence in the predictive ability of the model and contributes to its robustness in real-world situations.

Depending on the type of problem they are facing, these algorithms can be regression or classification algorithms like the ones in this case. The main difference between a regression and classification problem lies in the nature of the task being tackled in the field of machine learning. In a regression problem, the goal is to predict a continuous numerical value, such as a home's price or temperature. In this context, the models are trained to understand the relationship between the input variables and the target variable, seeking to generate predictions that are as close as possible to the actual values.

On the other hand, in a classification problem, the goal is to assign a label or category to a given instance. For example, classifying emails as spam or non-spam, predicting whether or not an accident will occur, etc. Here, the models learn patterns that allow the input to be assigned to one of several predefined categories. That is, while regression focuses on predicting numerical quantities, classification focuses on assigning labels to discrete data.

Fig. 5. Modeling Overview

| Problem                        | Possibility of incident                                                                                                                                                                                                                                                                                                                | Type of incident                                                                                                                                                                                                                                                                                               |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question                       | Will there be an incident?                                                                                                                                                                                                                                                                                                             | What type of incident?                                                                                                                                                                                                                                                                                         |
| Dataset                        | VesselBalancedSample (50% Involved, 50% Not involved)                                                                                                                                                                                                                                                                                  | MergedActivity (Incidents)                                                                                                                                                                                                                                                                                     |
| Response                       | · involved: Yes / No                                                                                                                                                                                                                                                                                                                   | · event_class: Critical Events / Onboard Emergencies / Maritime Accidents / Material Issues / Third-party Damages                                                                                                                                                                                              |
| Type of response               | Binary classification                                                                                                                                                                                                                                                                                                                  | Multilevel classification (5)                                                                                                                                                                                                                                                                                  |
| Characteristics<br>(Variables) | · vessel_class<br>· build_year<br>· gross_ton<br>· vessel_length<br>· flag_abbr<br>· classification_soc<br>· solas_desc                                                                                                                                                                                                                | · activity_id<br>· hour<br>· region<br>· longitude<br>· water_type<br>· damage_status<br>· vessel_class<br>· age<br>· gross_ton<br>· vessel_length<br>· air_temp<br>· wind_speed                                                                                                                               |
| Models<br>(Algorithms)         | 1. Naïve Bayes (NB)<br>2. Bagged Naïve Bayes -Gaussian- (BNB)<br>3. Gradient Boosting Machine (GBM)<br>4. Xtreme Gradient Boosting (XGB)<br>5. Random Forest (RF)<br>6. Support Vector Machine -Radial Kernel- (SVM)<br>7. Multilayer Perceptron (MLP)<br>8. Classification and Regression Trees (CART)<br>9. Logistic Regression (LR) | 1. Naïve Bayes (NB)<br>2. Gradient Boosting Machine (GBM)<br>3. Random Forest (RF)<br>4. Multilayer Perceptron (MLP)<br>5. Classification and Regression Trees (CART)<br>6. Keras: Sequential Model (Densely-connected Model)<br>7. Keras: Functional API (Multi-input)<br>8. Extra: H2o Auto Machine Learning |
| Evaluation<br>(Metrics)        | 1. ROC AUC<br>2. Accuracy<br>3. Log loss<br>4. Kappa<br>5. Matthews Correlation Coefficient (MCC)<br>6. F1<br>7. Sensitivity<br>8. Specificity                                                                                                                                                                                         | 1. ROC AUC<br>2. Accuracy<br>3. Log loss<br>4. Kappa<br>5. Matthews Correlation Coefficient (MCC)<br>6. F1<br>7. Recall<br>8. Precision                                                                                                                                                                        |

Fig. 5. Modeling summary

## 5.0. Theoretical framework

### 5.0.1. Machine learning models

To put in context and explain how they work, the following is going to be presented in the most summarized way possible, the operation that underpins the models and their techniques. The next section will look at the metrics used to compare with each other. In this analysis, the following models have been used:

#### 1. Naïve Bayes

Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem. It assumes conditional independence between the given characteristics of the classes, which simplifies the calculation of probabilities. Use Bayes' rule to calculate the probability of belonging to a given class an observation. The calculation of the conditional probability of belonging to a class  $C$  given a set of characteristics  $X$  is formulated with the following formula:

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

Where  $P(X|C)$  is the probability of observing  $X$  given that it belongs to class  $C$ ,  $P(C)$  is the prior probability of class  $C$ , and  $P(X)$  is the marginal probability of given  $X$ .

#### 2. Tree Augmented Naive Bayes Classifier (TAN):

TAN improves Naïve Bayes by allowing dependencies between features, represented by a tree. The structure of the tree is learned during training, capturing conditional dependency relationships between features and improving accuracy in cases where Naïve Bayes assumes independence. In this case, the joint probability is expressed as:

$$P(X, C) = P(C) \cdot \prod_{i=1}^n P(X_i|C, P_a(X_i))$$

Where  $P_a(X_i)$  are  $X_i$ 's parent features in the tree.

#### 3. Tree Augmented Naive Bayes Classifier Structure Learner Wrapper (TAN S):

It is a variant of TAN that uses search techniques to find the best tree structure, improving the learning efficiency and the quality of the resulting model, optimizing the joint probability  $P(X, C)$

#### 4. Hill Climbing Tree Augmented Naive Bayes Classifier (TAN Hill Climbing):

Similar to TAN, but uses the "hill climbing" algorithm to look for the tree structure that maximizes the probability of the observed data. It searches for the best solution locally, iteratively exploring close neighbors and moving in the direction that improves the target value, iteratively improving the structure of the tree.

#### 5. Aggregating One-Dependence Estimators Naive Bayes Classifier (AODE):

AODE is an extension of Naïve Bayes that allows dependencies between features, but assumes that these dependencies can be added independently to simplify the calculation, but assumes that these dependencies can be added independently:

$$P(X, C) = P(C) \cdot \prod_{i=1}^n P(X_i|C)$$

## 6. Gradient Boosting:

Gradient Boosting encompasses a set of algorithms that build predictive models in the form of decision trees sequentially, each correcting the errors of the previous model. It is optimized by minimizing a loss function, minimizing the loss function  $L$  iteratively by fitting the  $F$  model so that:

$$F_m(x) = F_{m-1}(x) + p \cdot h_m(x)$$

Where  $h_m(x)$  is the weak model fitted in the  $m$ -th iteration, and  $p$  is the learning rate.

## 7. Extreme Gradient Boosting (XGBTree):

XGBoost is an efficient implementation of Gradient Boosting. XGBTree uses decision trees as base models, incorporating regularization and pruning techniques to prevent overfitting. The target function to be minimized is as follows:

$$\sum_{i=1}^n L(y_i, F(x_i)) + \sum_{k=1}^K \Omega(f_k)$$

Where  $L$  is the loss function and  $\Omega(f_k)$  is the complexity penalty of the  $k$ -tree.

## 8. Random Forest:

The Random Forest algorithm builds multiple decision trees during training and combines their predictions for a more robust and accurate output. Instead of relying on a single tree, Random Forest creates a set of trees, each trained on a random sample of data and features. It then combines the predictions from these trees to improve generalization and reduce overfitting, providing a more robust and effective model.

## 9. Support Vector Machines (svmRadial):

Support Vector Machines are classification algorithms that seek to find the hyperplane that best separates classes in feature space. svmRadial uses a radial kernel to handle nonlinear datasets. The radial (Gaussian) kernel in the svmRadial algorithm is defined as:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

Where  $\sigma$  is a kernel width parameter

## 10. Multilayer perceptron:

The multilayer perceptron is a type of neural network with a hidden layer. Neural networks consist of layers of nodes (neurons) connected to each other, where each connection has a weight. Training involves adjusting these weights to minimize the difference between predictions and actual goals. In a neural network, the output of the  $j$ -layer is computed as:

$$y_j = f\left(\sum_{i=1}^n w_{ij} \cdot x_i + b_j\right)$$

Where  $w_{ij}$  is the weights,  $x_i$  is the inputs,  $b_j$  is the bias, and  $f$  is the trigger function of the model.

#### 11. **C5.0 Tree:**

C5.0 is a decision tree construction algorithm that uses information gain and pruning techniques to create a predictive model. It is efficient and effective for large data sets. Information gain is calculated as the reduction in entropy.

#### 12. **Logistic regression:**

Logistic regression is a regression model used for binary classification problems. Use the logistics function to model the probability of class membership. This function can be formulated as follows:

$$P(Y = 1|X) = \frac{1}{1 + \exp(-\beta_0 - \beta_1 \cdot X_1 - \dots - \beta_n \cdot X_n)}$$

Where  $\beta_0, \beta_1, \dots, \beta_n$  are the coefficients to be learned.

#### 13. **Densely populated network**

A densely populated neural network, or fully connected neural network, is a type of neural network architecture in which each neuron in one layer is connected to all the neurons in the next layer. These networks are commonly used in deep learning tasks. The output of a layer is calculated as exposed for the neural network. In the case of this analysis, the implementation of keras tensorflow will be used.

The mentioned models will be trained with the frameworks: skelarn, keras and h2o. To facilitate handling, all training sessions were exported to external files, thus being accessible for use for other purposes.

To give coherence to the explanatory nature of the models, comparative graphs of the relative importance of the variables have been used in the first instance, presenting individual results as well as together.

On the other hand, in order to compare their performance, the same metrics and methodology have been calculated for all of them. For certain models such as Keras or h2o, ad-hoc functions have had to be built to follow a process that is as consistent as possible. Next, let's see which ones have been used in this analysis.

### 5.0.2. Metrics

Evaluation metrics in machine learning are essential tools for measuring and understanding the performance of classification models. Each of the metrics provides insight into different aspects of the model. This makes it possible to compare the performance of the models and to be able to choose the best ones. The most commonly used for classification models such as those used in this analysis are explained below.

#### 1. ROC AUC

The Receiver Operator Curve (ROC) is a graphical representation of the sensitivity to the false positive rate for different classification thresholds. It provides a complete view of a model's performance at different levels of sensitivity and specificity. On the other hand, the AUC (Area Under Curve) measures the overall discriminative capacity of the model and is calculated as the area under the ROC curve. An AUC close to 1 indicates good performance, while an AUC of 0.5 indicates random performance. It can be calculated as:

$$AUC = \int_0^1 TPR(FPR)dx$$

Where TPR is the rate of true positives and FPR is the rate of false positives at each point.

#### 2. Accuracy

Measure the total proportion of instances correctly classified by the model. It helps to select models with an appropriate balance between sensitivity and specificity and to compare classifier quality in different contexts. It can be calculated using the following formula:

$$Accuracy = \frac{True\ Positives + True\ Negatives}{Total\ of\ Instances}$$

#### 3. Log Loss

Also known as logarithmic loss or cross-entropy loss, is a metric used to evaluate the accuracy of a classification model that outputs probabilities. It measures the uncertainty of the predictions by penalizing incorrect probability estimates more severely. A lower Log Loss indicates a better model.

$$Log\ Loss\ (multiclass) = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log(p_{ij})$$

Where:  $M$  is the number of classes.  $y_{ij}$  is 1 if instance  $i$  belongs to class  $j$ , otherwise 0. And  $p_{ij}$  is the predicted probability of instance  $i$  being in class  $j$ .

#### 4. Kappa

The kappa statistic adjusts observed accuracy to randomly expected accuracy and measures the agreement between predictions and observations. It is useful when class imbalance affects the interpretation of accuracy. It is calculated as follows:

$$k = \frac{Observed\ accuracy - Random\ Expected\ Accuracy}{1 - Random\ Expected\ Accuracy}$$

A kappa value of 1 indicates perfect match, 0 indicates random agreement, and negative values suggest worse than random agreement.

### 5. Matthews Correlation Coefficient (MCC)

It takes into account true positives, true negatives, false positives, and false negatives, providing a balanced measure that can be used even with imbalanced datasets. MCC values range from -1 to 1, where 1 indicates a perfect prediction, 0 indicates no better than random prediction, and -1 indicates total disagreement between prediction and observation. It can be calculated as follows:

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Where: *TP* is the number of true positives. *TN* is the number of true negatives. *FP* is the number of false positives. *FN* is the number of false negatives.

### 6. F1

It is the harmonic mean of precision and recall, providing a balance between the two. The F1 score ranges from 0 to 1, where 1 indicates perfect precision and recall, and 0 indicates the worst performance.

$$F1 \text{ Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

### 7. Sensitivity and Specificity

Sensitivity measures the proportion of positive instances that are correctly identified by the model. That is, a rate of true positives. Specificity, on the other hand, measures the proportion of negative instances that are correctly identified by the model. That is, a rate of true negatives. They are useful when there is asymmetry in the importance of type I (false positives) and type II (false negatives) errors. They can be calculated as follows:

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

### 8. Recall and Precision

The equivalent of above but for multiclass models, calculated as:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

On the other hand, it should be noted that in addition to these evaluation metrics, it is crucial to consider reliability, interpretability, training and prediction time, scalability, handling of missing data, computational cost, error tolerance, continuous learning, ethics and bias, and documentation/maintenance when evaluating machine learning models.

Next, the results of the training and evaluation of each of the models will be presented. On the one hand, those referring to the classification problem of this analysis "Will there be an incident or not"? and separately those corresponding to What type of incident?

## 5.1. Possibility of incident: VesselBalancedSample

The code contained in Annex 5.1 has been used for the modelling of this set, and it is not necessary to balance or complete the sample to apply the models. The training method can be described as follows::

**Technology stack:** for the training of the models of this binary classification problem, the *sklearn* library has been used.

**Data partitioning:** 80% of observations for training and 20% for testing. It has been divided with a random selection.

**Hyperparameter tuning:** Each model has been adjusted seeking to maximize the performance metrics, testing the values by executing successive training sessions guided by the graphs obtained. They have been established as final after verifying minimal variations in the metrics.

**Evaluation Metric:** ROC (Receiver Operating Characteristic)

As for the results, this report will only highlight the comparison of models, since the details of the parameters used in each model and their specific outputs can be found in the aforementioned annex.

### 5.1.1. Comparison of the “Possibility of incident” models

The main metrics of the models proposed for this problem will be compared. All of them have been trained with the sklearn framework

| Model performance comparison for training sample |          |           |        |        |        |         |        |              |              |
|--------------------------------------------------|----------|-----------|--------|--------|--------|---------|--------|--------------|--------------|
|                                                  | ROC AUC: | Accuracy: | Kappa: | RMSE:  | MAE:   | R2:     | F1:    | Sensitivity: | Specificity: |
| <b>RF</b>                                        | 0.9515   | 0.9515    | 0.903  | 0.0485 | 0.0485 | 0.8059  | 0.9525 | 0.9717       | 0.9312       |
| <b>CART</b>                                      | 0.9515   | 0.9515    | 0.903  | 0.0485 | 0.0485 | 0.806   | 0.9523 | 0.9663       | 0.9367       |
| <b>GBM</b>                                       | 0.8662   | 0.8662    | 0.7324 | 0.1338 | 0.1338 | 0.465   | 0.8732 | 0.9195       | 0.8128       |
| <b>NNET</b>                                      | 0.8638   | 0.8639    | 0.7277 | 0.1361 | 0.1361 | 0.4555  | 0.8716 | 0.9232       | 0.8044       |
| <b>XGB</b>                                       | 0.8491   | 0.8492    | 0.6984 | 0.1508 | 0.1508 | 0.3969  | 0.8591 | 0.9179       | 0.7804       |
| <b>SVM</b>                                       | 0.8479   | 0.848     | 0.696  | 0.152  | 0.152  | 0.3922  | 0.8613 | 0.9423       | 0.7536       |
| <b>RL</b>                                        | 0.838    | 0.8381    | 0.6761 | 0.1619 | 0.1619 | 0.3523  | 0.8517 | 0.9285       | 0.7474       |
| <b>BNB</b>                                       | 0.6734   | 0.6731    | 0.3466 | 0.3269 | 0.3269 | -0.3077 | 0.5821 | 0.4547       | 0.892        |
| <b>NB</b>                                        | 0.6733   | 0.673     | 0.3463 | 0.327  | 0.327  | -0.3082 | 0.5821 | 0.455        | 0.8915       |

Model performance comparison for test sample

|             | ROC AUC: | Accuracy: | Kappa: | RMSE:  | MAE:   | R2:     | F1:    | Sensitivity: | Specificity: |
|-------------|----------|-----------|--------|--------|--------|---------|--------|--------------|--------------|
| <b>RF</b>   | 0.883    | 0.8828    | 0.7657 | 0.1172 | 0.1172 | 0.5311  | 0.8866 | 0.9213       | 0.8447       |
| <b>CART</b> | 0.879    | 0.8789    | 0.7578 | 0.1211 | 0.1211 | 0.5155  | 0.8818 | 0.9088       | 0.8493       |
| <b>GBM</b>  | 0.8651   | 0.8648    | 0.7299 | 0.1352 | 0.1352 | 0.4594  | 0.871  | 0.9176       | 0.8127       |
| <b>NNET</b> | 0.8577   | 0.8574    | 0.7149 | 0.1426 | 0.1426 | 0.4295  | 0.8645 | 0.9153       | 0.8001       |
| <b>XGB</b>  | 0.8515   | 0.8511    | 0.7024 | 0.1489 | 0.1489 | 0.4044  | 0.8597 | 0.9175       | 0.7855       |
| <b>SVM</b>  | 0.8492   | 0.8487    | 0.6977 | 0.1513 | 0.1513 | 0.3947  | 0.8609 | 0.9416       | 0.7569       |
| <b>RL</b>   | 0.8426   | 0.8421    | 0.6845 | 0.1579 | 0.1579 | 0.3683  | 0.8542 | 0.9301       | 0.755        |
| <b>BNB</b>  | 0.6738   | 0.6751    | 0.3486 | 0.3249 | 0.3249 | -0.2996 | 0.5799 | 0.451        | 0.8967       |
| <b>NB</b>   | 0.6737   | 0.6749    | 0.3482 | 0.3251 | 0.3251 | -0.3003 | 0.58   | 0.4514       | 0.8959       |

The 12 models used in this analysis have good performance for both the test and validation samples, so they could be used to predict the possibility of an incident in the naval field. Graphically overall:

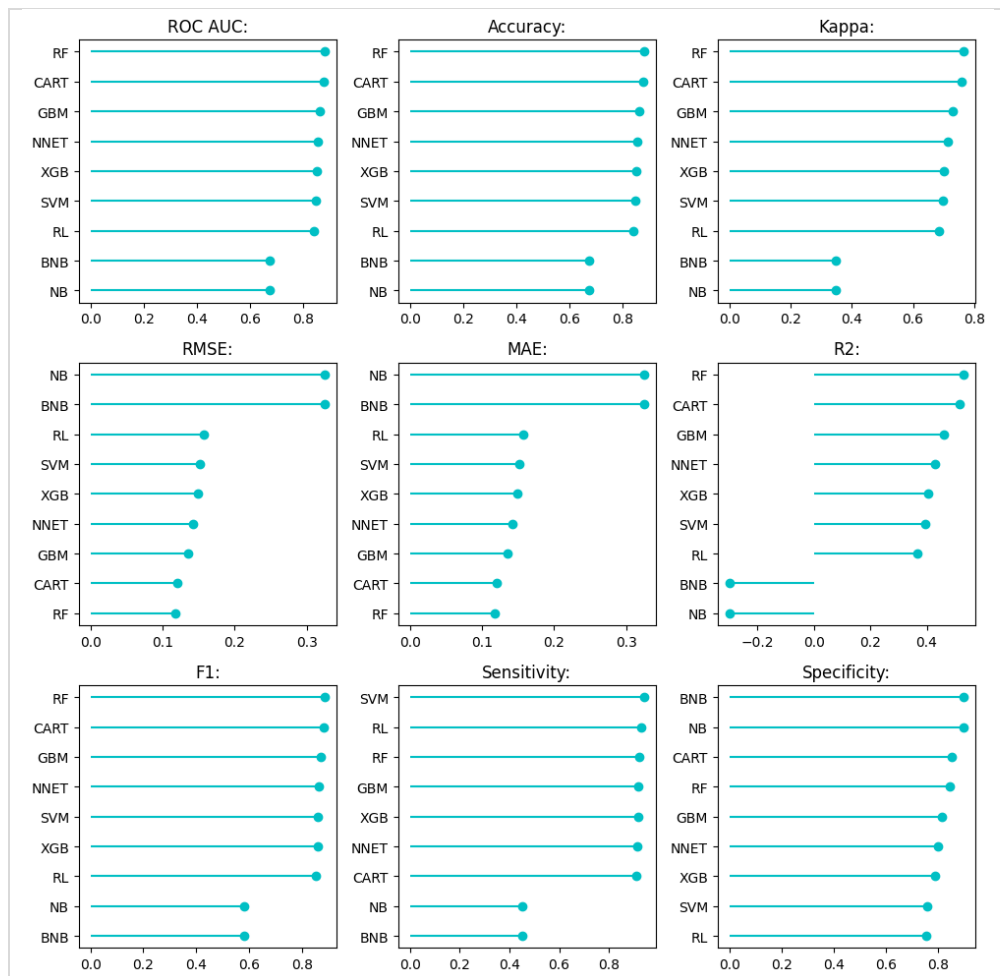


Fig. 5.1.1.1. Metrics comparison

And reviewing a comparison of the ROC curve for the main models:

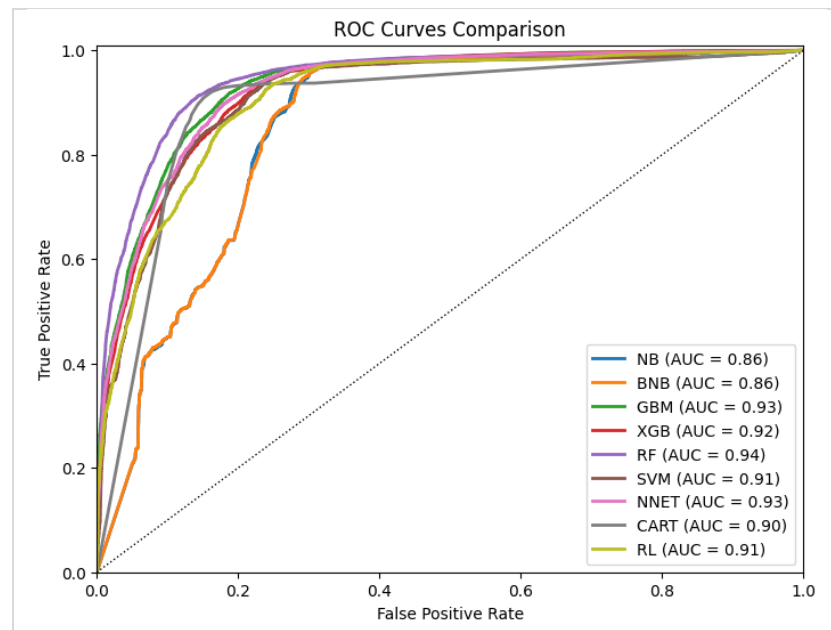


Fig. 5.1.1.2. ROC comparison

You can see that they follow a very similar distribution. Regarding the metrics of each model, the following particular statements can be considered:

- **Random Forest (RF):**

It stands out with the highest AUC (0.883) and an impressive accuracy of 88.2%. In addition, it has a good balance between sensitivity (92.1%) and specificity (84.4%). The Kappa coefficient of 0.7657 indicates substantial agreement between the model's predictions and the actual rankings.

- **Classification And Regression Tree (CART):**

Similar to GBM in many ways, the decision tree achieves an AUC of 0.879 and an accuracy of 87.8%. Sensitivity (90.8%) and specificity (84.9%) are consistent with good classification performance.

- **Gradient Boosting (GBM):**

Although slightly below Random Forest, GBM shows a solid performance with an AUC of 0.865 and an accuracy of 86.4%. Sensitivity (91.76%) and specificity (81.2%) suggest that it is capable of managing both classes effectively.

- **Support Vector Machine (SVM):**

Although SVM shows the highest specificity (94.1%), its overall accuracy is lower (84.8%). This suggests that it is more cautious when classifying the positive class, which may be beneficial depending on the context.

- **Multilayer Perceptron (MLP):**

Despite having a slightly lower accuracy (91.5%), MLP demonstrates a high sensitivity of 80.0%, indicating a robust ability to identify the positive class. The AUC of 0.857 suggests a good performance in class discrimination.

- **Naive Bayes (NB):**

It shows weak performance with an AUC of 0.673 and an accuracy of 67.4%. The sensitivity (45.1%) is too low compared to other models, suggesting that it may have difficulty identifying the positive class.

In the context of marine incident analysis, the importance of sensitivity versus specificity in machine learning models is directly related to the critical nature and potential serious consequences of marine incidents. The following aspects can be considered:

- Critical Nature of Marine Incidents:

Incidents in the marine environment, such as collisions, groundings, or spills of hazardous substances, can have significant impacts on the safety of navigation, the marine environment, and human life. The model's ability to correctly detect these events is crucial.

- Sensitivity

Sensitivity refers to the model's ability to correctly identify positive cases, i.e., actual incidents. In the marine context, high sensitivity means that the model is effective in detecting most incidents, which is essential to ensure a rapid and effective response.

- Specificity

Specificity refers to the model's ability to correctly identify negative cases, i.e., normal, uneventful situations. Although important, in the marine environment, specificity might be less critical compared to sensitivity. The reason is that false negatives (undetected incidents) could have more serious consequences than false positives (incorrect alarms).

- Costs Associated with False Negatives:

In the marine field, missing a real incident (false negative) can have disastrous consequences in terms of safety and the environment. For example, failing to detect a spill of hazardous substances in time could lead to delayed emergency responses, exacerbating the environmental impact.

- High Sensitivity Preference:

Since the priority in the marine realm is usually early and accurate detection of incidents, a model with a higher sensitivity is likely to be preferred, even if this leads to a slightly lower specificity. The ability to identify the vast majority of incidents can help minimize risk and associated consequences.

- Adaptation to Variable Marine Conditions:

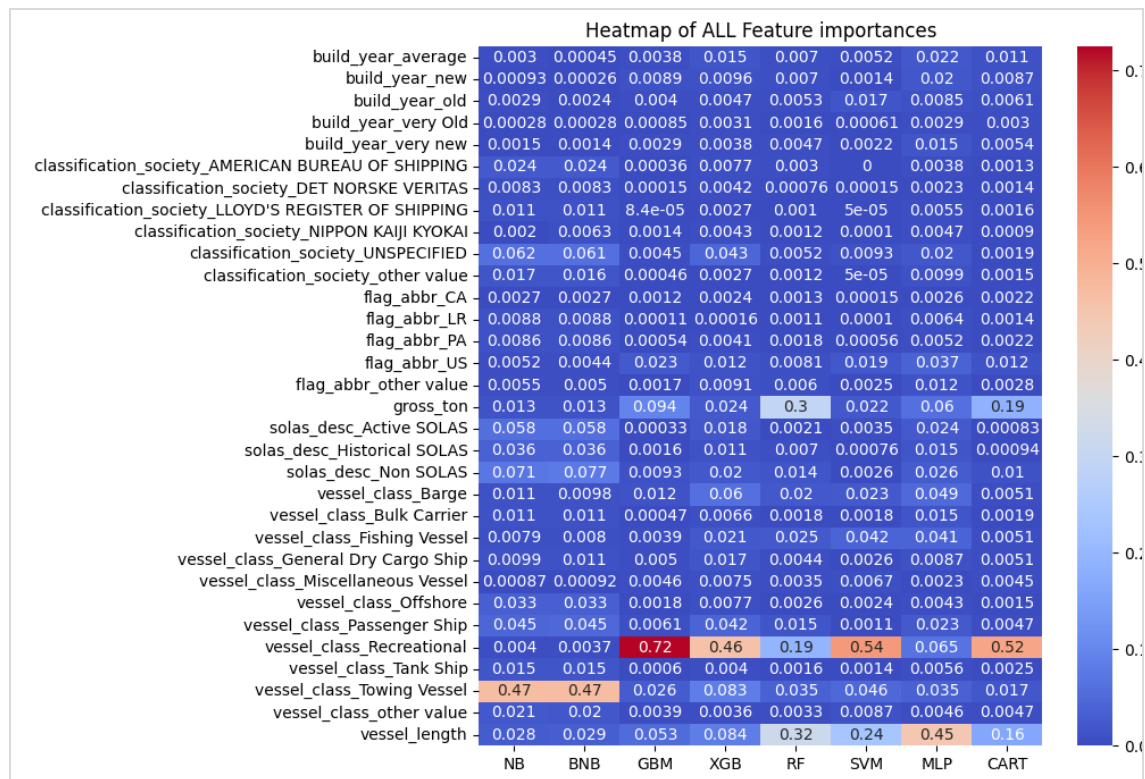
Marine conditions can be variable and challenging, which increases the importance of having a robust model that can adapt to various situations and detect incidents in different scenarios.

In marine incident analysis, sensitivity tends to be more critical than specificity. A model that maximizes sensitivity, albeit at the expense of specificity, might be preferred to ensure effective incident detection and thus contribute to marine safety and environmental protection.

There are several models that are in this situation, such as Random Forest, or Extreme Gradient Boosting that offer good performance. The fact is that Random Forest has slightly higher evaluation metrics, is considered to be more robust and resistant to over-fitting. In addition, its interpretability is simpler and it is very efficient in terms of training time, so it could be considered the best option for the resolution of this classification problem.

### 5.1.2. Explainability of "Possibility of Incident" Models

In this case, we are going to focus the analysis on the importance of the variables through a heat map to more easily detect common behaviors in the models:



· Fig. 5.1.2.1. Feature Importances

#### Key Observations:

- vessel\_class\_Recreational: This feature shows high importance across all models, particularly in GBM\_train and XGB\_train.
- vessel\_length: This feature also shows notable importance, especially in rf\_train and cart\_train.
- vessel\_class\_Passenger Ship: This feature is relatively important in all models, with rf\_train giving it the most importance.

#### Less Important Features:

- Features like flag\_abbr\_CA, classification\_society\_LLOYD'S REGISTER OF SHIPPING, and solas\_desc\_Historical SOLAS have low importance across all models.

#### Model Differences:

- Random Forest and CART seem to give more weight to vessel\_length compared to the other models.
- GBM and XGB have similar patterns in feature importance, highlighting vessel\_class\_Recreational as significantly important.

The heatmap shows that while some features like vessel\_class\_Recreational and vessel\_length are universally important, others vary in importance depending on the model.

On the other hand, we can analyse SHAP Values (SHapley Additive exPlanations). SHAP values provide a game theory-consistent explanation of the contribution of each feature to the prediction.

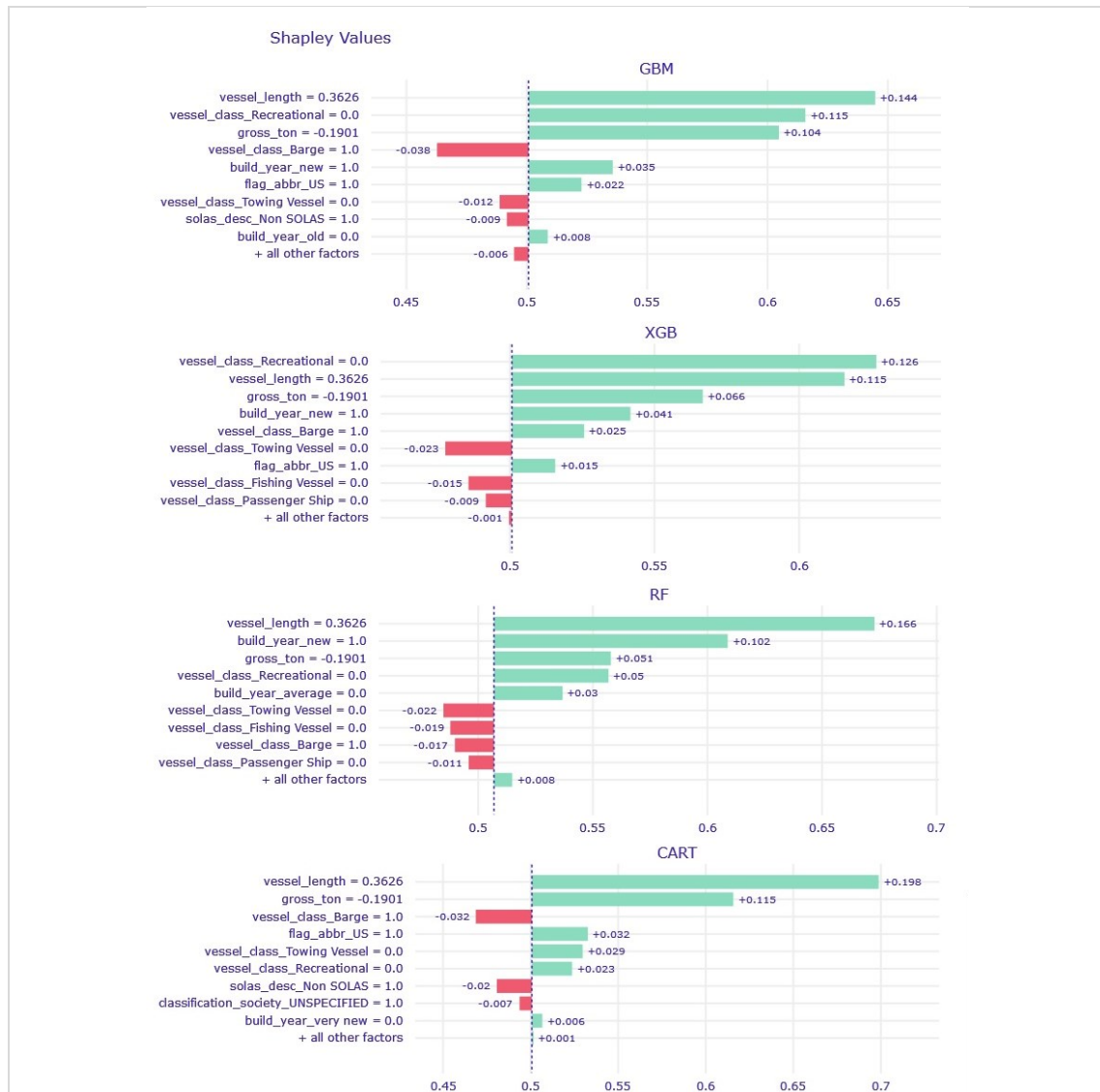


Fig. 5.1.2.2. SHAP values

#### Consistent Top Feature:

- vessel\_length: Highly important and positively influential across all models.
- Some features have a mixed impact depending on the model, like vessel\_class\_Recreational, which shows varying levels of importance and direction.

#### Key Features per Model:

- GBM: vessel\_length, vessel\_class\_Recreational (both positive).
- XGB: vessel\_length, vessel\_class\_Recreational (both positive), gross\_ton (negative).
- RF: vessel\_length, build\_year\_new (both positive).
- CART: vessel\_length (positive), gross\_ton (negative).

#### Negative Impact Features:

- gross\_ton: Negative impact in GBM, XGB, and CART.
- Various vessel classes like vessel\_class\_Towing Vessel and vessel\_class\_Fishing Vessel show negative impacts in RF.

## 5.2. Incident Type: MergedActivity

The code in Annex 5.2 has been used for modelling in this dataset. In this case, it has been necessary to balance the sample and fill in the missing values, according to the following methods:

### 5.2.0. Data Cleansing: Balancing and Handling Missing Values

Target variable, *event\_class*, presents the following imbalance in its values before applying the predictive models:

|               |                        |                           |                        |                            |                            |
|---------------|------------------------|---------------------------|------------------------|----------------------------|----------------------------|
| Variable:     | <i>Critical Events</i> | <i>Maritime Accidents</i> | <i>Material Issues</i> | <i>Onboard Emergencies</i> | <i>Third-party Damages</i> |
| Observations: | 16938                  | 18467                     | 17158                  | 6630                       | 8807                       |

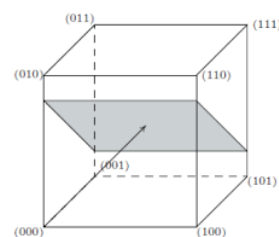
Since algorithms need a balanced variable to optimize their operation, it has been decided to determine 9000 observations per level of the variable to make up the final sample. According to this figure, 26% synthetic observations should be generated for the underrepresented class (Onboard Emergencies)

For subsampling, the **Cube method** has been chosen for Critical-Events, Maritime.Accidents and Material.Issues. It is a technique used in the field of data mining and signal processing to reduce the amount of data in a set so that a meaningful representation of the original information is maintained. To do this, this method selects samples so that the Horvitz-Thompson estimators are equal to the totals of the known auxiliary variables, which helps to balance the sample.

It consists of two phases, according to the method proposed by Deville and Tillé (2004):

#### 1. Flight Phase:

It can be conceptualized as a random path that starts at a  $\pi$  inclusion probability vector and moves through a feature space. This random path remains at the intersection between the cube and the subspace defined by the equilibrium equations  $Q$ .



#### 2. Landing Phase:

At the end of the flight phase, if a vertex (sample) has not been selected, the landing phase is activated. There are three possible solutions for the landing phase:

- Progressively Eliminate Equilibrium Variables:** Equilibrium variables are progressively eliminated and the flight phase is reapplied. It is necessary to suppress the variables in order from least to most important.

b. Linear Programming for Approximately Balanced Sample. Linear programming is used to calculate the best roughly balanced sample. The goal is to minimize the difference in balance between the categories.

c. Choosing the Vertex Closest to the Probability Vector The vertex closest to the probability vector obtained in the flight phase is selected. Inclusion probabilities that are not yet equal to 0 or 1 are rounded.

#### Efficient Implementation

Deville and Tillé developed a faster implementation of the flight phase, detailed in "Fast SAS Macros for balancing simple user's guide." This implementation contributes to the efficiency of the method and offers advantages, such as having no restrictions on the size of the population. The execution time is linearly dependent on the size of the population, which improves the scalability of the method.

This method is commonly applied in situations where the data set is too large to process efficiently or when trying to balance disproportionate classes, as is the case here

For oversampling, the **SMOTE** method is applied to Onboard.Emergencies, Third-party.Damages. There are other methods such as ROS and ADASYN, but for this analysis this one has been chosen because of its good performance and wide general acceptance for this task.

SMOTE (Synthetic Minority Over-sampling Technique) aims to address the problem of unbalanced classes by generating synthetic instances of minority classes. Two main phases can be distinguished:

#### 1. Instance Selection:

For instances to be oversampled,  $k$  close neighbors are selected (usually  $k=5$  is chosen). Neighbors are selected using some distance metric, such as Euclidean distance.

#### 2. Generation Process:

For each feature in the instance, SMOTE calculates the difference between the value of the feature in the original instance and the corresponding value in one of the neighbors. Multiply this difference by a random number between 0 and 1 and add it to the original value of the feature to generate the value of the new synthetic instance.

This method operates in feature space, creating synthetic instances that form a line in feature space between the original instance and its neighbors. This helps expand the space occupied by the minority class and improves the model's ability to generalize correctly.

SMOTE has several variants, such as SMOTE-NC (for datasets with categorical characteristics), Borderline-SMOTE (focuses on instances close to the boundary between classes)

After the application of both techniques, a balanced sample has been obtained. However, there are missing values of which the treatment is explained below.

Regarding the treatment of missing values, in this data set, three variables have been detected with this problem to a greater or lesser extent:

|                 |                 |                   |                          |
|-----------------|-----------------|-------------------|--------------------------|
| Variable:       | <i>air_temp</i> | <i>wind_speed</i> | <i>damage_assessment</i> |
| Missing values: | 2030            | 5513              | 15                       |

Missing values can lead to the introduction of biases in the estimation of models as they distort variability and generalization. Therefore, avoiding a treatment of missing values has a direct impact on accuracy and performance. And even many machine learning algorithms won't even start the training phase if this problem is declared.

In order to fix this, the used method has been MICE. MICE involves imputing missing values in a dataset through a series of chained equations, where each variable with missing data is imputed conditional on the other variables in the dataset. The process iteratively cycles through each variable that has missing values, imputing these values based on a specified model (such as linear regression, logistic regression, etc.) that includes all other variables as predictors.

As a comparison of the quality of imputation, here is a table for means and standard deviations before and after applying the imputation:

|                                | <b>air_temp</b> | <b>wind_speed</b> | <b>damage_assessment</b> |
|--------------------------------|-----------------|-------------------|--------------------------|
| <b>Mean before impute</b>      | 149.763586      | 50.583439         | 122959.803912            |
| <b>Mean after impute</b>       | 149.75645       | 50.760162         | 122959.803926            |
| <b>Difference for Mean (%)</b> | -0.0048%        | 0.3494%           | 0.0000%                  |
|                                | <b>air_temp</b> | <b>wind_speed</b> | <b>damage_assessment</b> |
| <b>Stds before impute</b>      | 94.999003       | 30.646614         | 3887042.365253           |
| <b>Stds after impute</b>       | 91.242978       | 24.480288         | 3885441.465189           |
| <b>Difference for Std (%)</b>  | -3.9538%        | -20.1207%         | -0.0412%                 |

Fig. 5.2.0.1. Imputation differences

The imputation process has slightly adjusted the means and standard deviations of the variables, with *wind\_speed* showing the most significant change in variability. Overall, the MICE imputation process seems to have effectively handled the missing data without significantly altering the statistical properties of the dataset.

**Means:** The very small percentage differences indicate that the imputed values are close to the original values, implying consistency.

**Standard Deviations:** The decreases in standard deviations, especially for *wind\_speed*, suggest a reduction in variability, which could be due to the imputation smoothing out some of the noise or extreme values.

Regarding the training method of the models:

**Technology stack:** for the training of the models of this classification problem, the *sklearn*, *keras tensorflow* and *h2o* libraries have been used.

**Data partitioning:** 80% of observations for training and 20% for testing. It has been divided with a random selection.

**Hyperparameter tuning:** Each model has been adjusted seeking to maximize the performance metrics, testing the values by executing successive training sessions guided by the graphs obtained. They have been established as final after verifying minimal variations in the metrics.

**Evaluation metric:** On this occasion, when working with a multiclass predictor, "logLoss" has been chosen. Log Loss penalizes incorrect predictions by assigning a higher penalty when the probability assigned to the correct class is low. This better reflects the model's confidence in its predictions.

As for the results, as in the previous case, in this report only the comparison of models will be highlighted, since the details of the parameters used in each model and their specific outputs can be found in the aforementioned annex. In this section, a smaller number of models have been considered, although with the aim of providing a diversity of tools, models processed with three libraries have been used, being a simple task to extrapolate the configurations to other models.

### 5.2.1. Comparison of the "Type of Incident" models

In this case, the models trained with sklearn and Keras tensorflow and the H2o will be compared together

| Model performance comparison for training samples |          |           |           |        |        |        |         |            |
|---------------------------------------------------|----------|-----------|-----------|--------|--------|--------|---------|------------|
|                                                   | ROC AUC: | Accuracy: | Log loss: | Kappa: | MCC:   | F1:    | Recall: | Precision: |
| <b>RF</b>                                         | 0.9588   | 0.8668    | 0.6305    | 0.8334 | 0.8338 | 0.8645 | 0.8665  | 0.8639     |
| <b>CART</b>                                       | 0.8989   | 0.8152    | 5.7058    | 0.7690 | 0.7697 | 0.8130 | 0.8150  | 0.8140     |
| <b>GBM</b>                                        | 0.8227   | 0.5238    | 1.1612    | 0.4047 | 0.4053 | 0.5221 | 0.5233  | 0.5238     |
| <b>NNET</b>                                       | 0.8204   | 0.5128    | 1.1424    | 0.3908 | 0.3938 | 0.5055 | 0.5121  | 0.5129     |
| <b>keras Func</b>                                 | 0.8118   | 0.5004    | 1.1674    | 0.3754 | 0.3768 | 0.5033 | 0.4999  | 0.5145     |
| <b>h2o</b>                                        | 0.8100   | 0.5033    | 1.1998    | 0.3790 | 0.3800 | 0.5011 | 0.5030  | 0.5034     |
| <b>keras Sec</b>                                  | 0.7724   | 0.4512    | 1.2733    | 0.3140 | 0.3188 | 0.4470 | 0.4508  | 0.4687     |
| <b>NB</b>                                         | 0.6657   | 0.3375    | 4.2296    | 0.1712 | 0.1888 | 0.2970 | 0.3366  | 0.3952     |

· Fig. 5.2.1.1 Model metrics for training samples

Now the returns achieved by predictive models are neither as good nor as similar to each other, as in the previous question. For the training set, there are no bad metrics in

| Model performance comparison for test samples |          |           |           |        |        |        |         |            |
|-----------------------------------------------|----------|-----------|-----------|--------|--------|--------|---------|------------|
|                                               | ROC AUC: | Accuracy: | Log loss: | Kappa: | MCC:   | F1:    | Recall: | Precision: |
| <b>RF</b>                                     | 0.8587   | 0.6427    | 1.1947    | 0.5532 | 0.5561 | 0.6376 | 0.6438  | 0.6458     |
| <b>h2o</b>                                    | 0.8077   | 0.4947    | 1.2072    | 0.3686 | 0.3696 | 0.4944 | 0.4961  | 0.4968     |
| <b>GBM</b>                                    | 0.7914   | 0.4832    | 1.2390    | 0.3541 | 0.3549 | 0.4823 | 0.4849  | 0.4835     |
| <b>NNET</b>                                   | 0.7754   | 0.4518    | 1.2952    | 0.3155 | 0.3187 | 0.4455 | 0.4548  | 0.4528     |
| <b>keras Func</b>                             | 0.7735   | 0.4491    | 1.2679    | 0.3115 | 0.3129 | 0.4535 | 0.4512  | 0.4645     |
| <b>keras Sec</b>                              | 0.7566   | 0.4326    | 1.3044    | 0.2909 | 0.2953 | 0.4295 | 0.4343  | 0.4503     |
| <b>CART</b>                                   | 0.7504   | 0.5904    | 14.2234   | 0.4879 | 0.4910 | 0.5832 | 0.5914  | 0.5904     |
| <b>NB</b>                                     | 0.6579   | 0.3252    | 4.3607    | 0.1592 | 0.1760 | 0.2853 | 0.3287  | 0.3792     |

· Fig. 5.2.1.2 Model metrics for test samples

The performance of each model for the test sample, would be as follows:

· **Random Forest (RF):**

Achieves the best performance in terms of ROC AUC (0.8587), indicating a high ability to discriminate between classes. Also leads in precision (0.6458) and MCC (0.5561), making it the most balanced model.

· **Auto Machine learning (h2o):**

Has the second-highest ROC AUC (0.8077), but relatively low accuracy and F1 score. Shows moderate log loss (1.2072), suggesting better calibration compared to other models except RF and GBM.

· **Gradient Boosting Machine (GBM):**

Demonstrates good performance with a ROC AUC of 0.7914 and a log loss of 1.2390, showing a reasonable balance between discrimination and calibration. Its precision and F1 metrics are average compared to other models.

· **Neural Network (NNET):**

Although it has a ROC AUC of 0.7754, its accuracy (0.4518) and F1 score (0.4455) are low. The log loss is relatively high (1.2952), suggesting potential calibration issues.

· **Keras Models (Functional and Sequential):**

Both models have similar performance with ROC AUC around 0.77 and accuracies close to 0.45. Their precision, F1, and other metrics do not particularly stand out compared to the previously mentioned models.

· **CART (Classification and Regression Tree):**

Has a moderate ROC AUC (0.7504) but surprisingly high accuracy (0.5904). The log loss is extremely high (14.2234), indicating significant calibration problems.

· **Naive Bayes (NB):**

Shows the worst overall performance, with a ROC AUC of 0.6579 and the lowest accuracy (0.3252). Its other metrics (F1, MCC, Recall, etc.) are the lowest among all compared models.

Now taking into account the test and training results, they can be compared to outline the overfitting they support.

- **Random Forest (RF):**

The performance of the RF model on the training sample is significantly better than that on the test sample. This suggests that the model could be overfitting to the training data, as there is a noticeable decrease in all metrics when evaluated on the test sample.

- **Auto Machine learning (h2o):**

The h2o model shows impressive consistency between training and testing data. The metrics show only small decreases from training to testing, suggesting that the model is not overfitted and has good generalization ability. Compared to other models, h2o offers a strong and reliable balance between fit and generalization.

- **Gradient Boosting Machine (GBM):**

GBM shows a slight decrease in performance when going from training to testing, which is expected, but the difference is not as pronounced as in RF and CART, suggesting a better balance between fit and generalization.

- **Neural Network (NNET):**

Similar to GBM, NNET shows an expected decrease in performance when going from training to testing. The differences are reasonable and suggest a model that generalizes relatively well.

- **Keras Models (Functional and Sequential):**

In the test, performance decreases but not drastically, suggesting good generalization ability compared to CART and RF.

- **CART (Classification and Regression Tree):**

CART also shows a noticeable decrease in its performance from the training sample to the test sample. The dramatic increase in the log loss of the test sample suggests that the model is severely overfitted.

- **Naive Bayes (NB):**

NB shows a minor decrease from training to testing, although its overall performance remains the lowest. This suggests that the model is not overfitting, but it is not very efficient either.

So, these models can be grouped in four categories:

- Overfitting: Random Forest (RF) and CART
- Better generalization: GBM and NNET
- Average performance: Keras models (Functional and Sequential)
- Poor but consistent performance: Naive Bayes (NB)

### 5.2.2. Explainability of the "Incident Type" models

For this part of our analysis, we'll be using a heat map to visualize the significance of different variables. This approach will help us more easily spot shared patterns and trends across the various models:

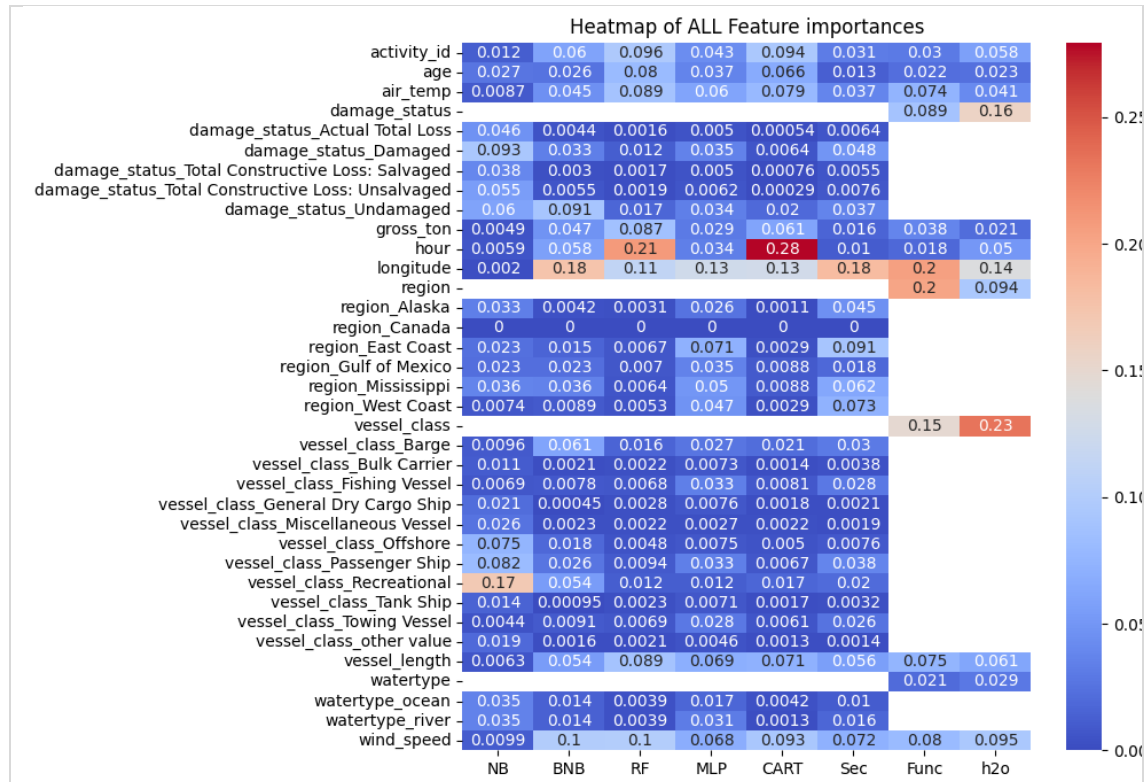


Fig. 5.2.2.1. Feature Importances

**gross\_tonn:** This variable is the most important across almost all models (RF, CART, h2o, Keras Func, Keras Sec, MLP).

**region\_Asia:** This variable is also highly important in several models (RF, h2o, Keras Func, Keras Sec, MLP, BNB).

**hour:** This variable stands out in models such as RF, h2o, and MLP.

**vessel\_class\_Barge:** Highly important in the CART model.

**Key Variables by Model:**

Random Forest (RF): gross\_tonn, region\_Asia, hour

CART: gross\_tonn, vessel\_class\_Barge, vessel\_class\_Tank Ship

h2o: gross\_tonn, region\_Asia, hour

Keras Func: gross\_tonn, region\_Asia, hour

Keras Sec: gross\_tonn, vessel\_class\_Bulk Carrier, region\_Asia

MLP: gross\_tonn, region\_Asia, hour

BNB: region\_Asia, gross\_tonn, region\_Canada

NB: gross\_tonn, region\_Canada, age

The variables gross\_tonn, region\_Asia, and hour are consistently important across multiple models, indicating their significant relevance in the dataset.

## 6. Conclusions

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The present study reveals the usefulness of the Random Forest model in the prediction of marine incidents, highlighting its ability to address complex datasets and capture nonlinear patterns effectively. One of the critical factors for the success of this analysis has been the extensive pre-processing of data, which included the pooling of information from various sources. This integration of data from multiple sites not only enriched the dataset, but also provided a more complete representation of marine conditions, thus improving the generalization of the model to real-world situations.

As explained, one of the keys to choosing a machine learning model in the field of marine incident prevention is prioritizing sensitivity over specificity. In this sense, the ability of the proposed model to manage data from different sites highlights its robustness and adaptability, essential aspects in the marine context, where geographic variability and the specific conditions of each region play a crucial role in the frequency of incidents.

The generalization of the model to real-world situations is supported by the diversity of data considered, allowing the model to learn more representative patterns and thus increasing its usefulness in predicting incidents in new circumstances.

In view of the results, it could be established that this analysis can provide a model that answers the first of the questions posed: Will a certain ship be involved in an incident? However, the second question in which one discerns what kind of incident it would be, at the moment, such a clear statement cannot be made. It remains for the future to propose other models and features that offer greater adjustments in this section.

There would also be another interesting question left for the future that has been left in the inkwell: How much would an incident cost to a certain ship? In this case, we would be facing a regression problem for the variable `damage_assessment`, which has an apparently good quality of data in the `MergedActivity` dataset.

However, the practical application of these results in maritime operations can be immediate, as the model's ability to predict incidents can significantly influence safety and operational decision-making. The advance information provided by the model allows for a proactive response to potential risks, which can result in a significant reduction in incidents and their consequences.

It is critical to recognize that this analysis is an iterative process, and the model is expected to benefit from continuous improvements. Constant feedback and updating the model with new data will contribute to its improvement and adaptation to changes in maritime conditions over time.

Finally, it is imperative to underline the importance of ethical and privacy considerations when implementing a predictive model in the maritime environment. Ensuring that the collection and use of data is done in an ethical manner and respectful of the privacy of the parties involved is essential for the sustained success and acceptance of the application of the model in operational environments.

## 7. Bibliography

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